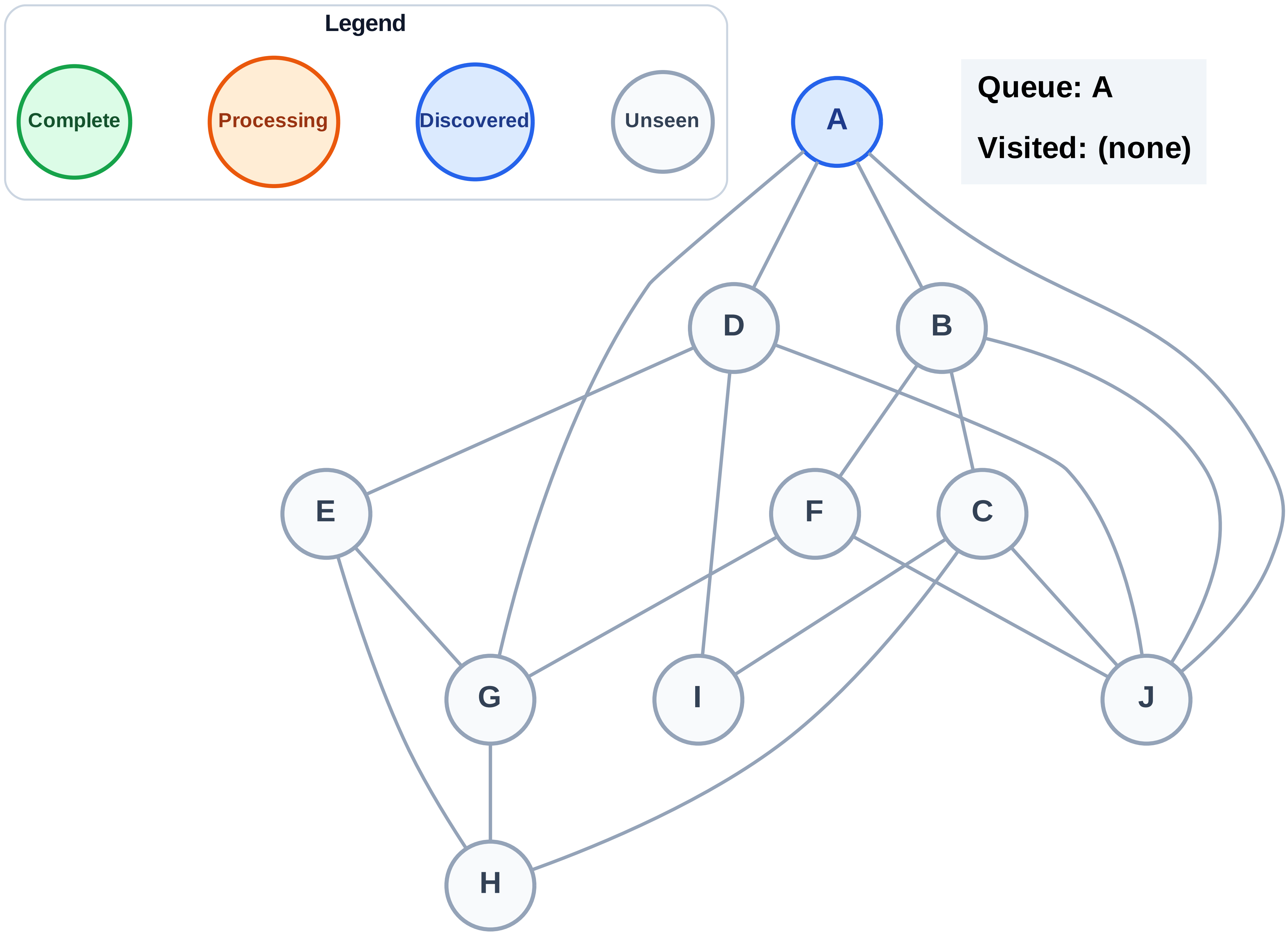
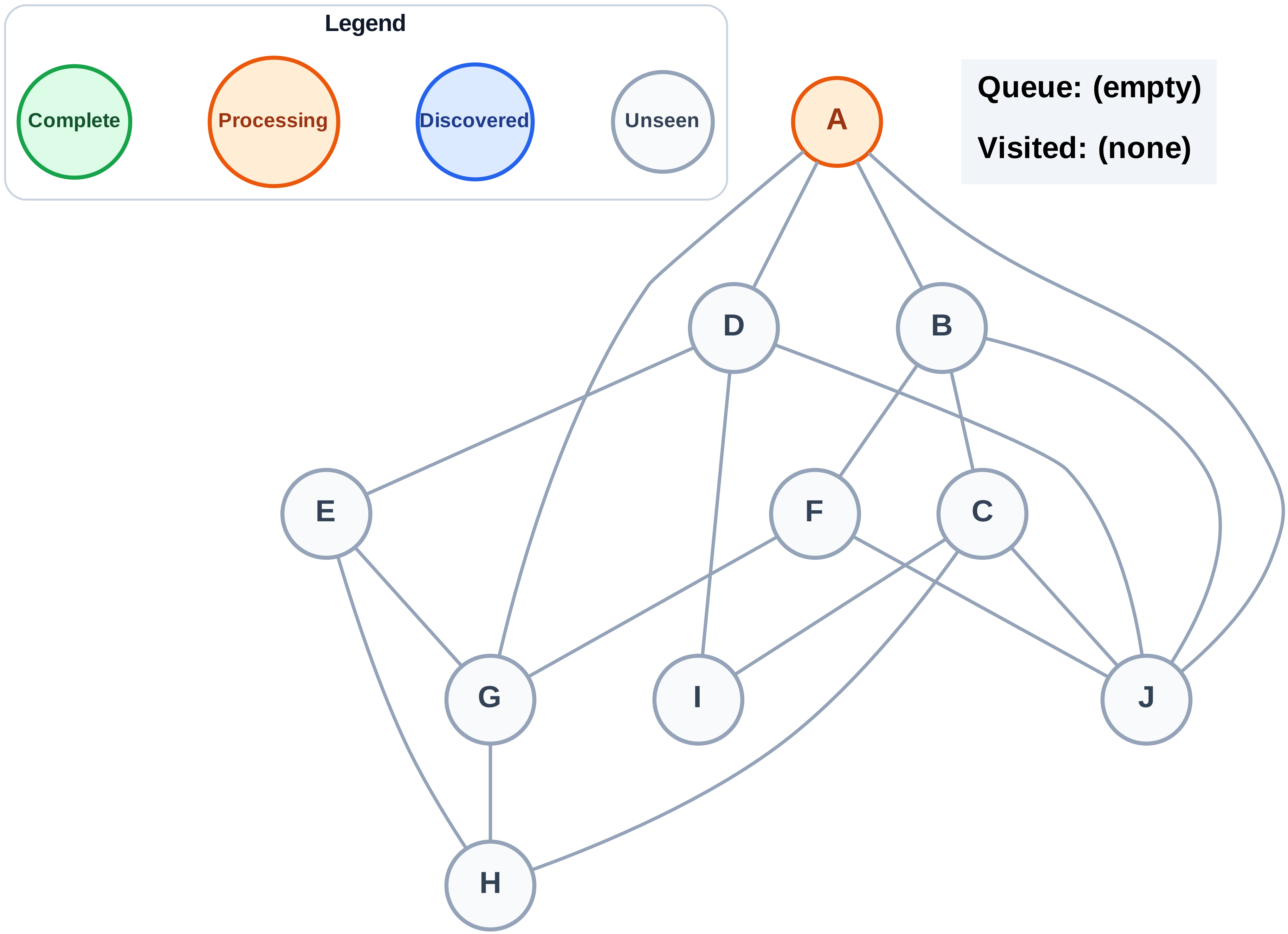
BFS Traversal

Step 1: Start at A



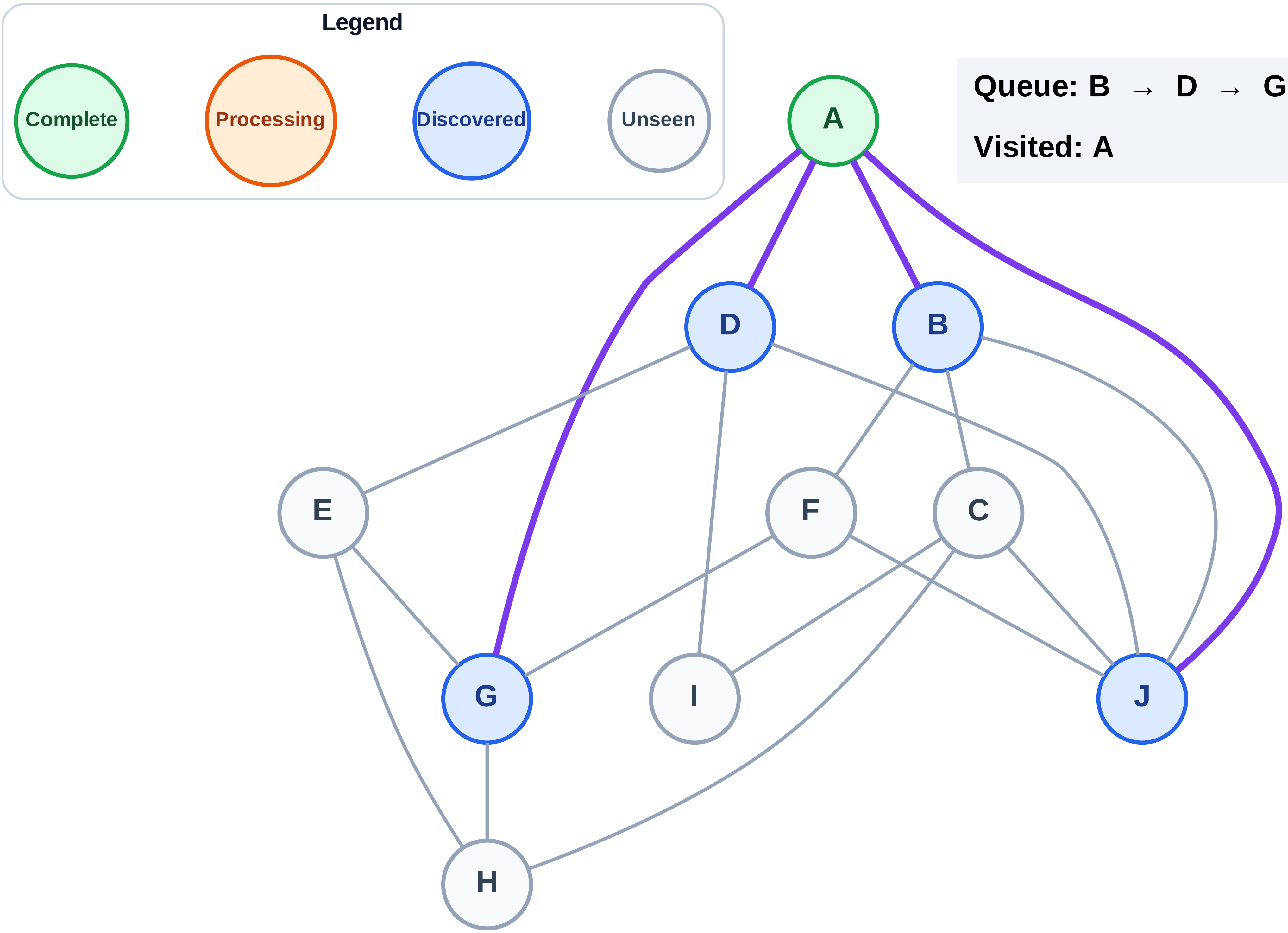
BFS Traversal

Step 2: Process node A



BFS Traversal

Step 3: Discover B, D, G, J from A

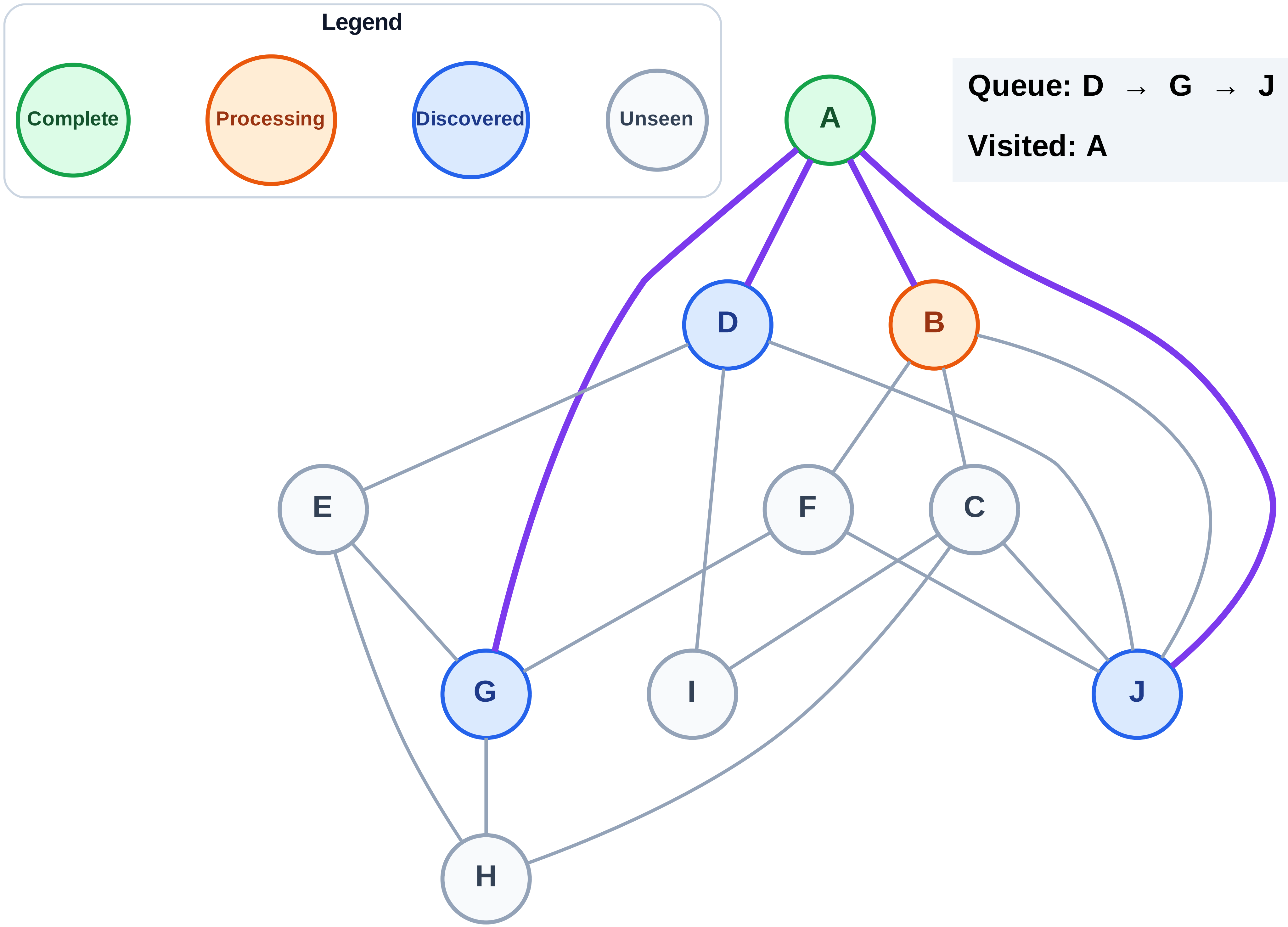


Queue: B → D → G → J

Visited: A

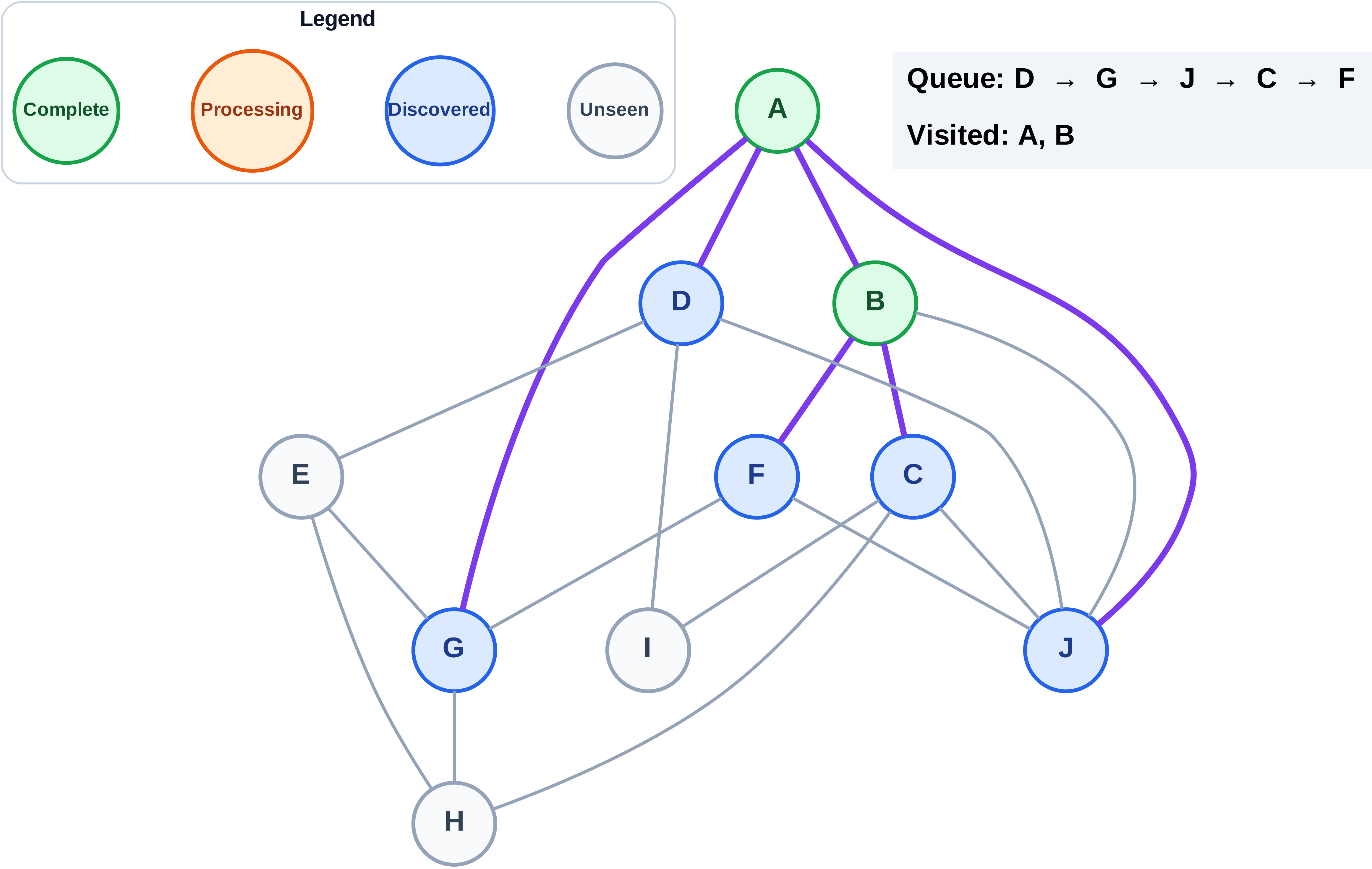
BFS Traversal

Step 4: Process node B



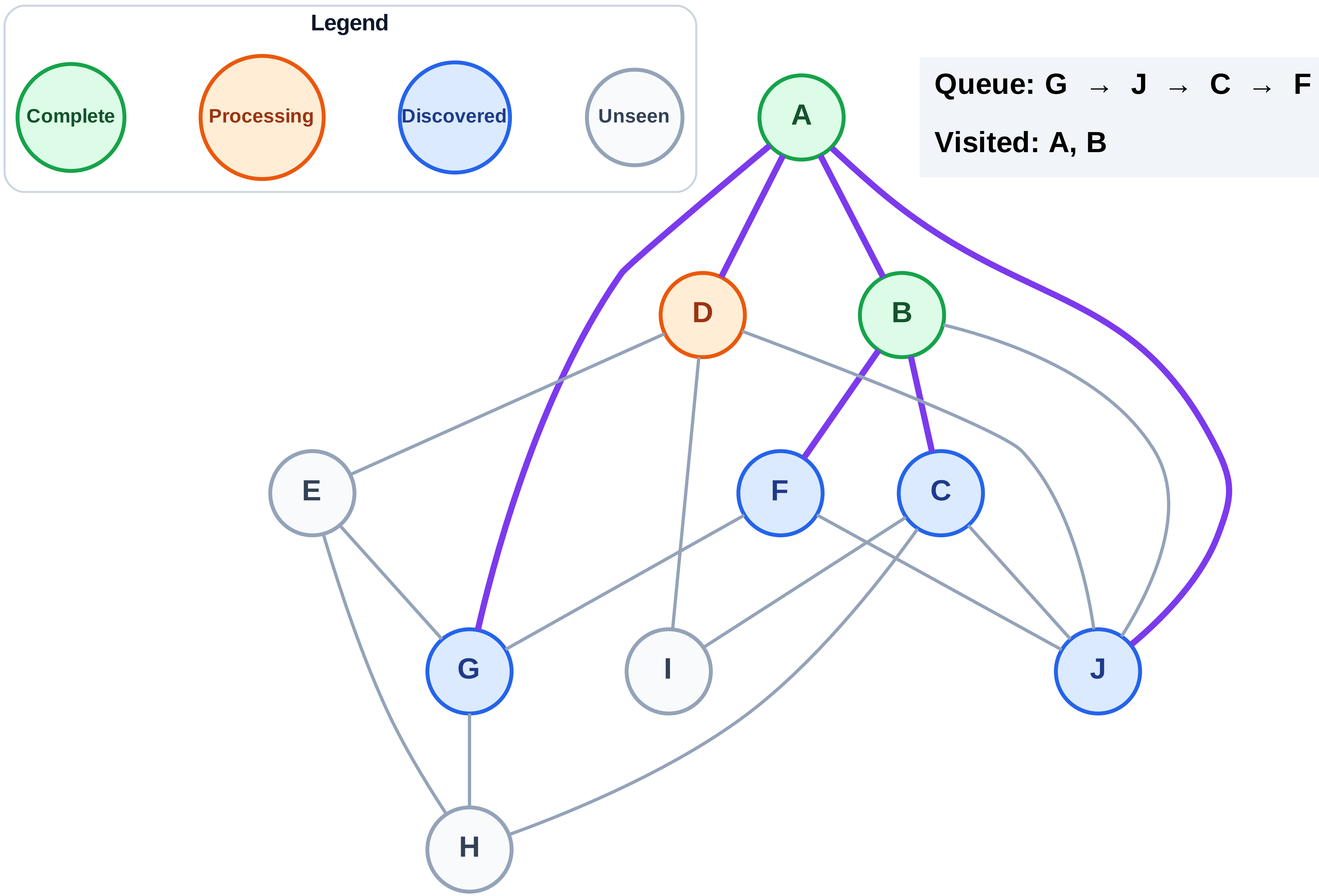
BFS Traversal

Step 5: Discover C, F from B



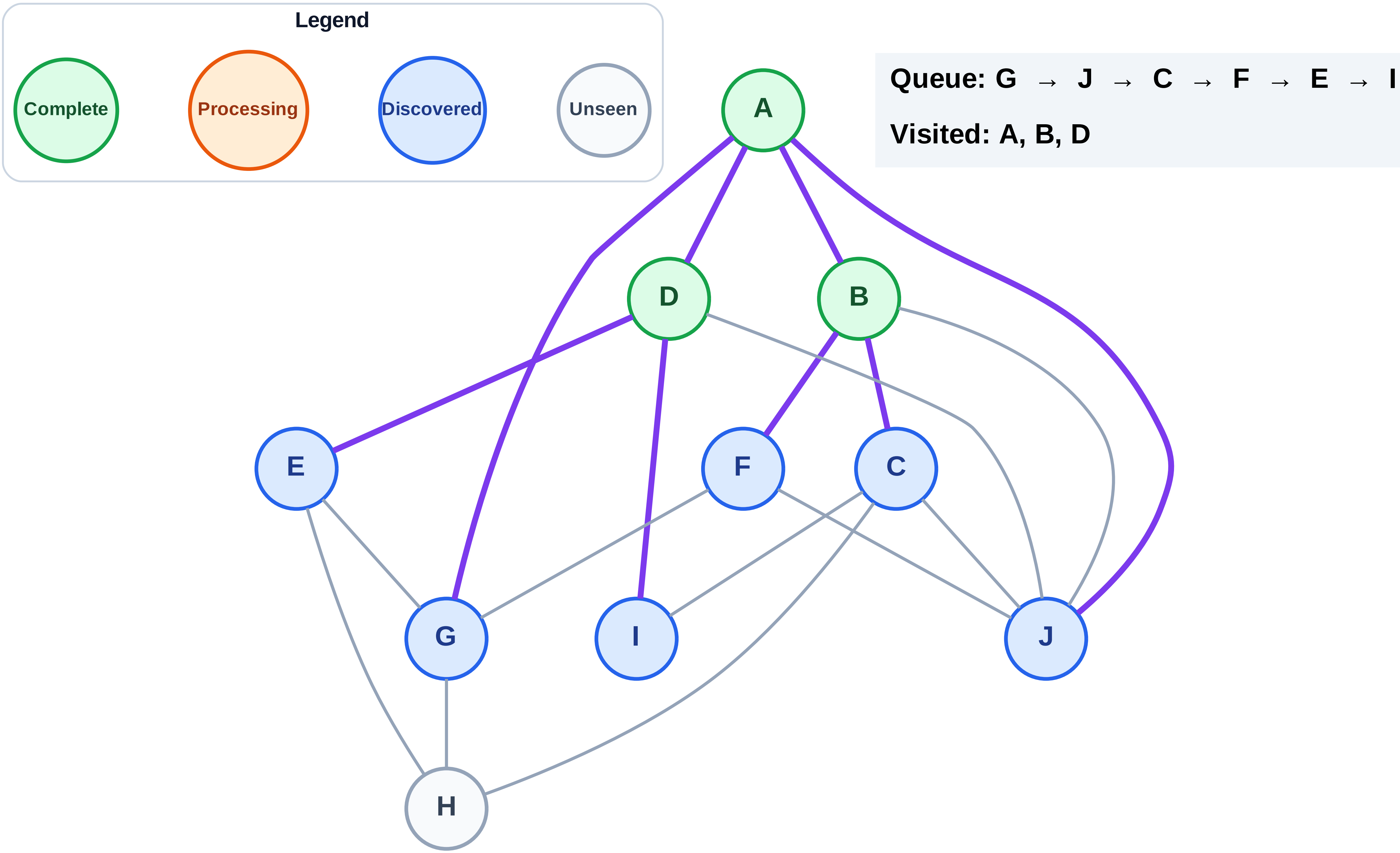
BFS Traversal

Step 6: Process node D



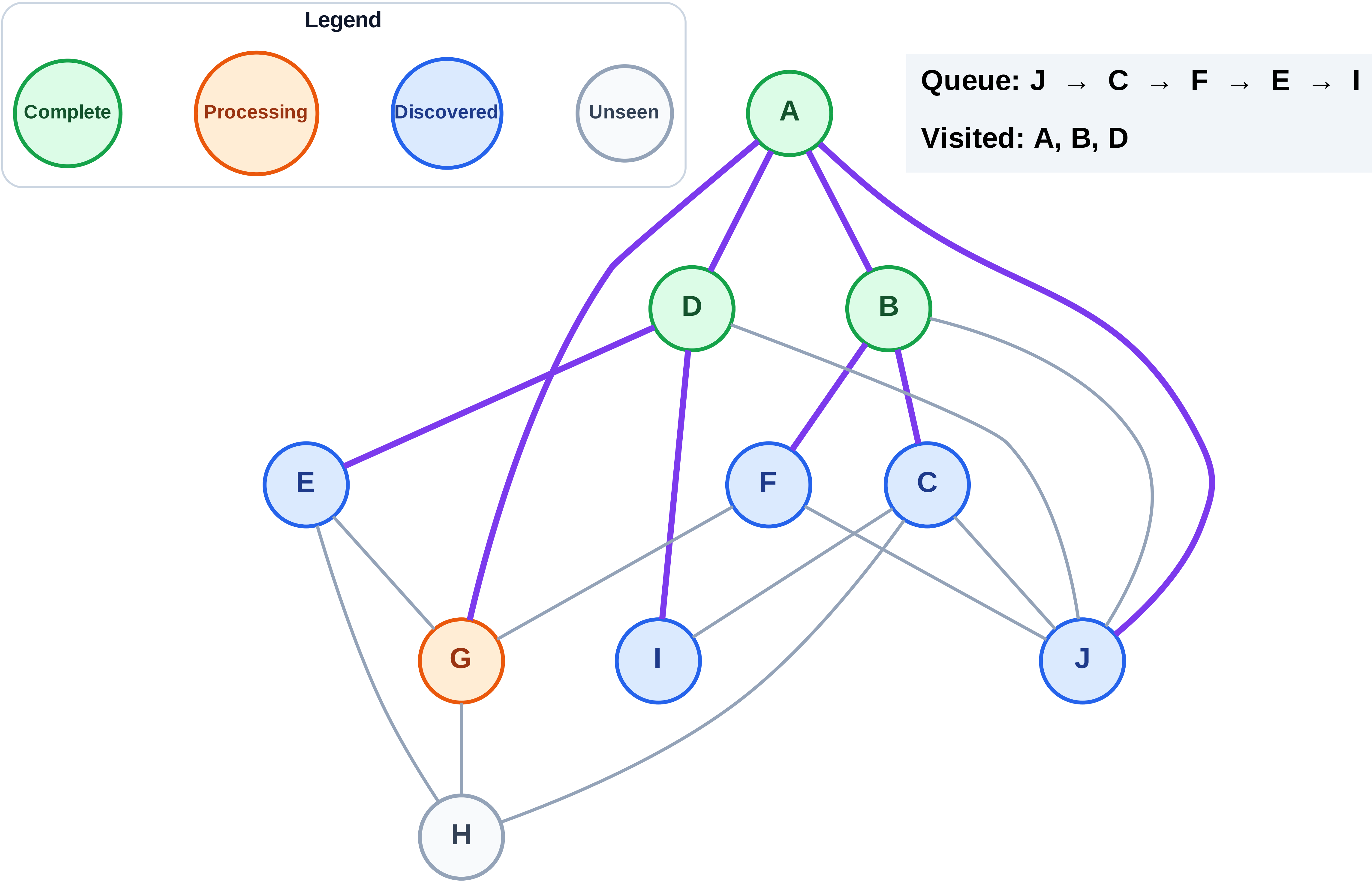
BFS Traversal

Step 7: Discover E, I from D



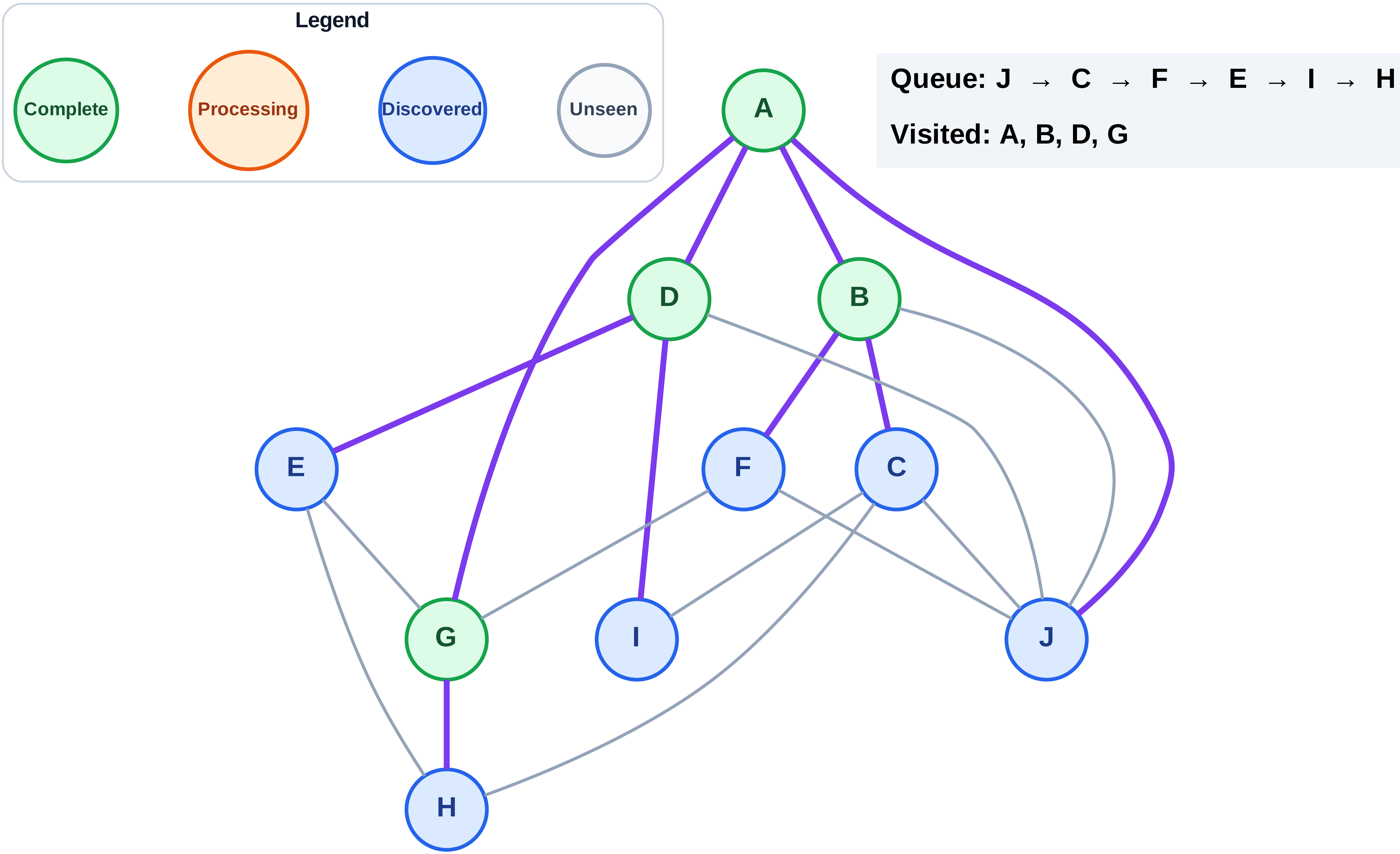
BFS Traversal

Step 8: Process node G



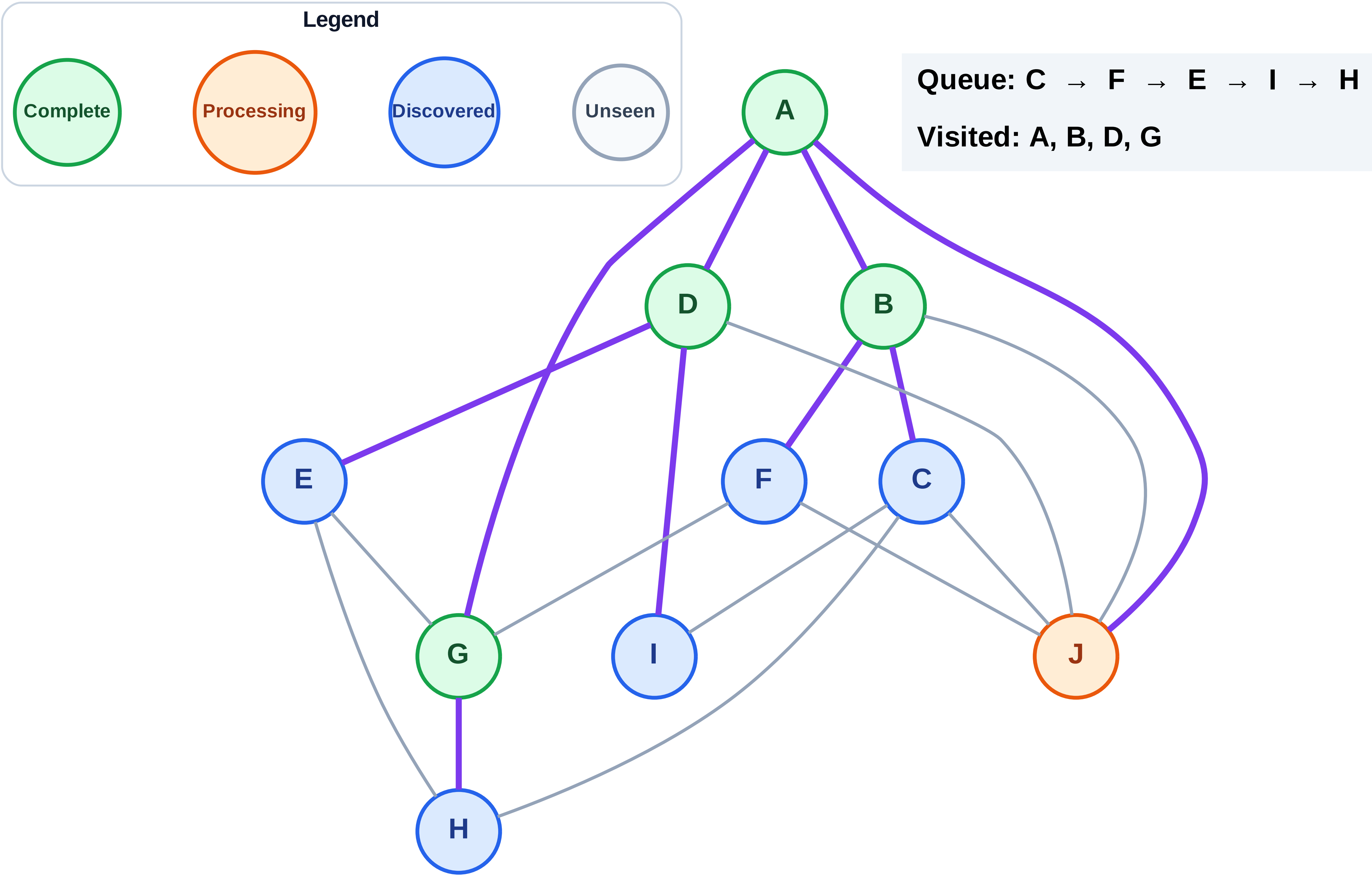
BFS Traversal

Step 9: Discover H from G



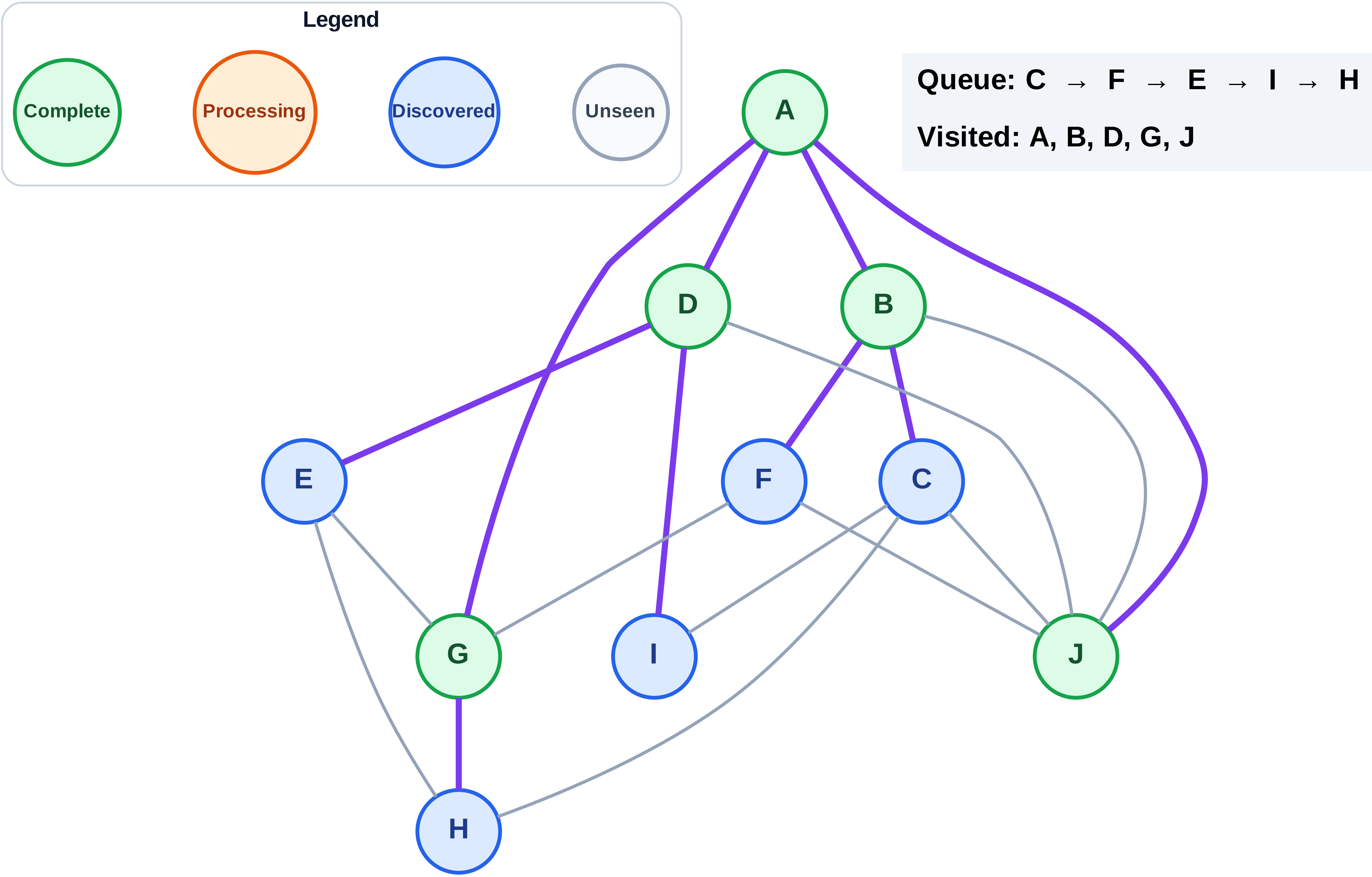
BFS Traversal

Step 10: Process node J



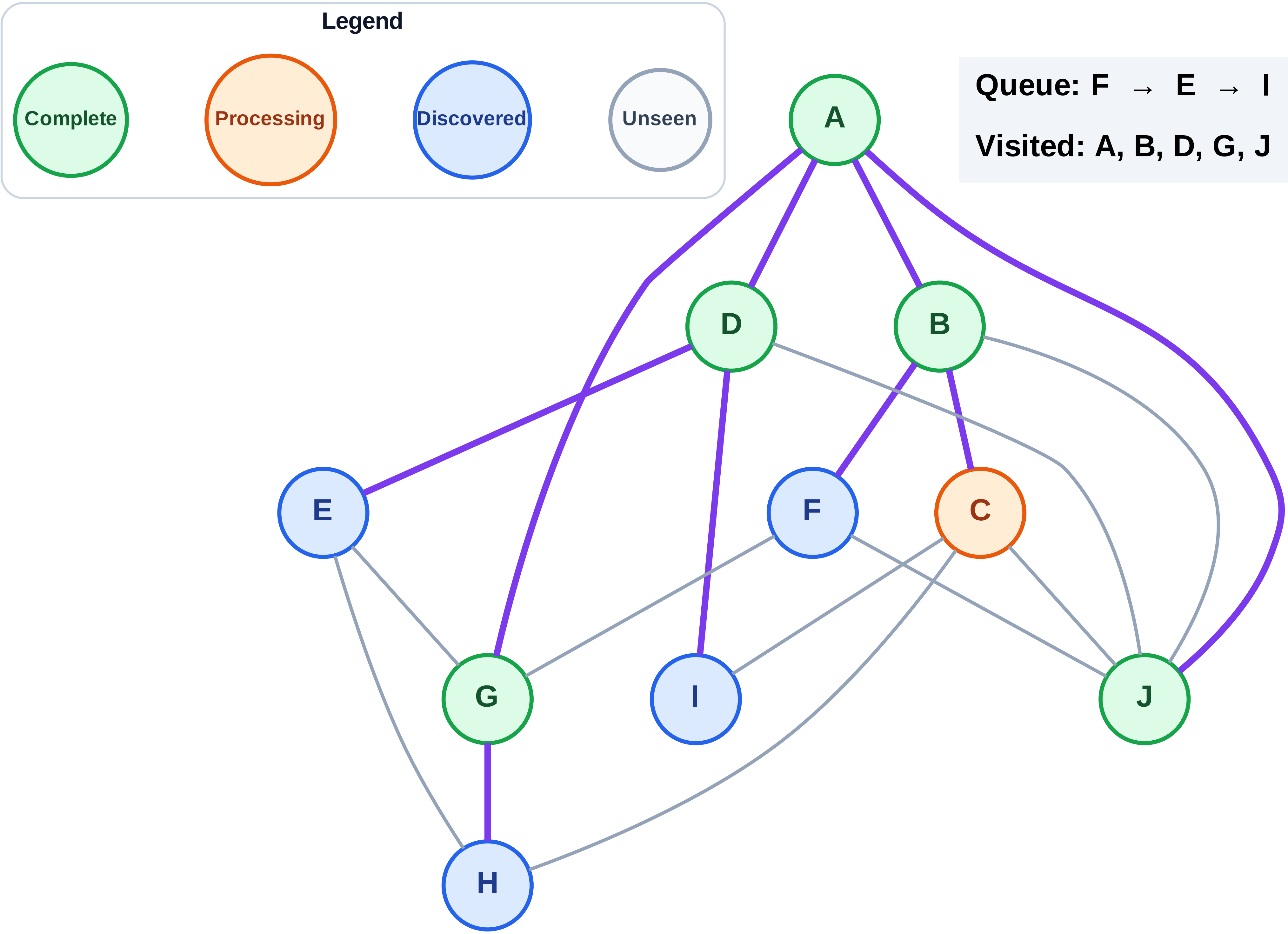
BFS Traversal

Step 11: No new neighbors from J



BFS Traversal

Step 12: Process node C

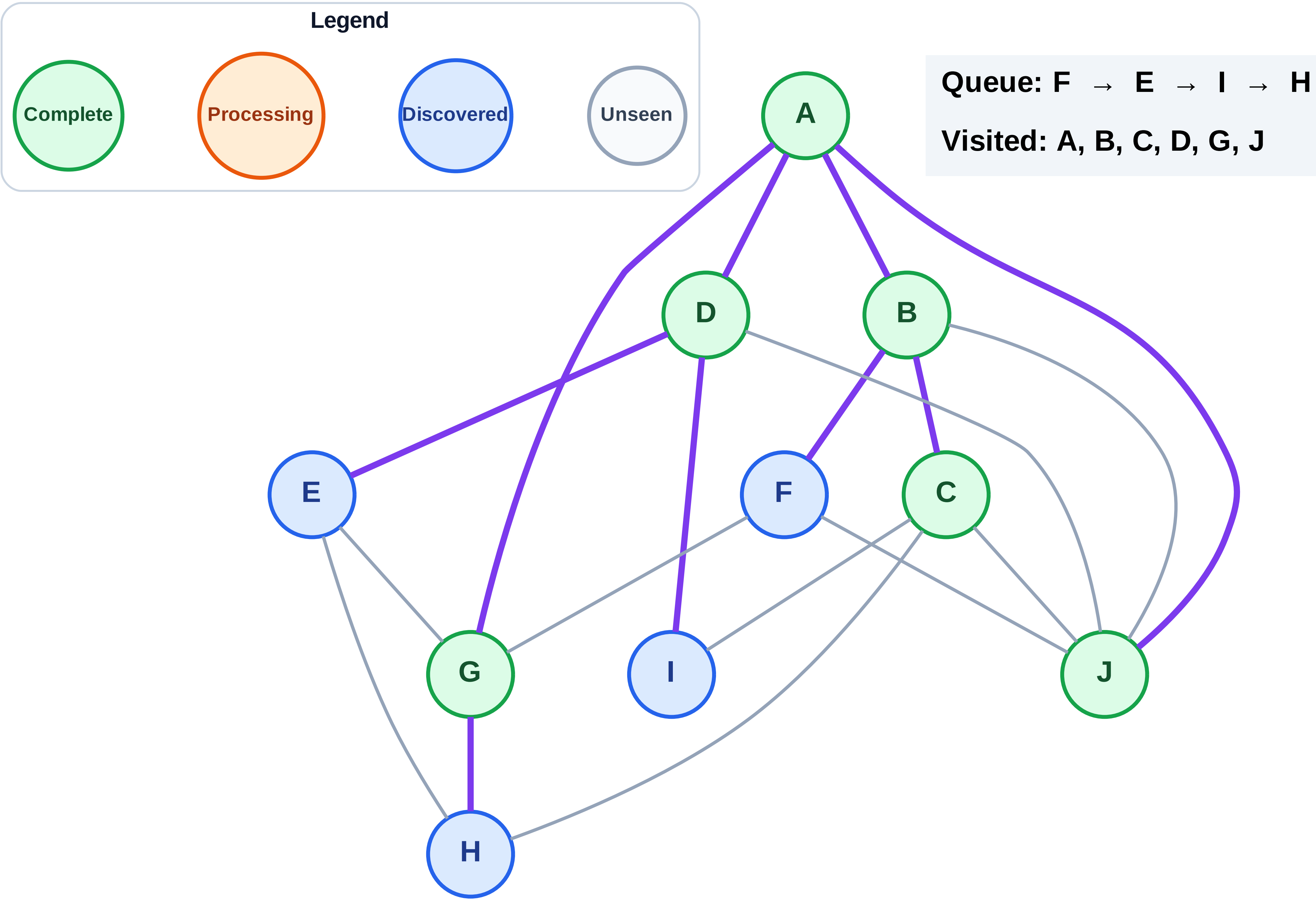


Queue: F → E → I → H

Visited: A, B, D, G, J

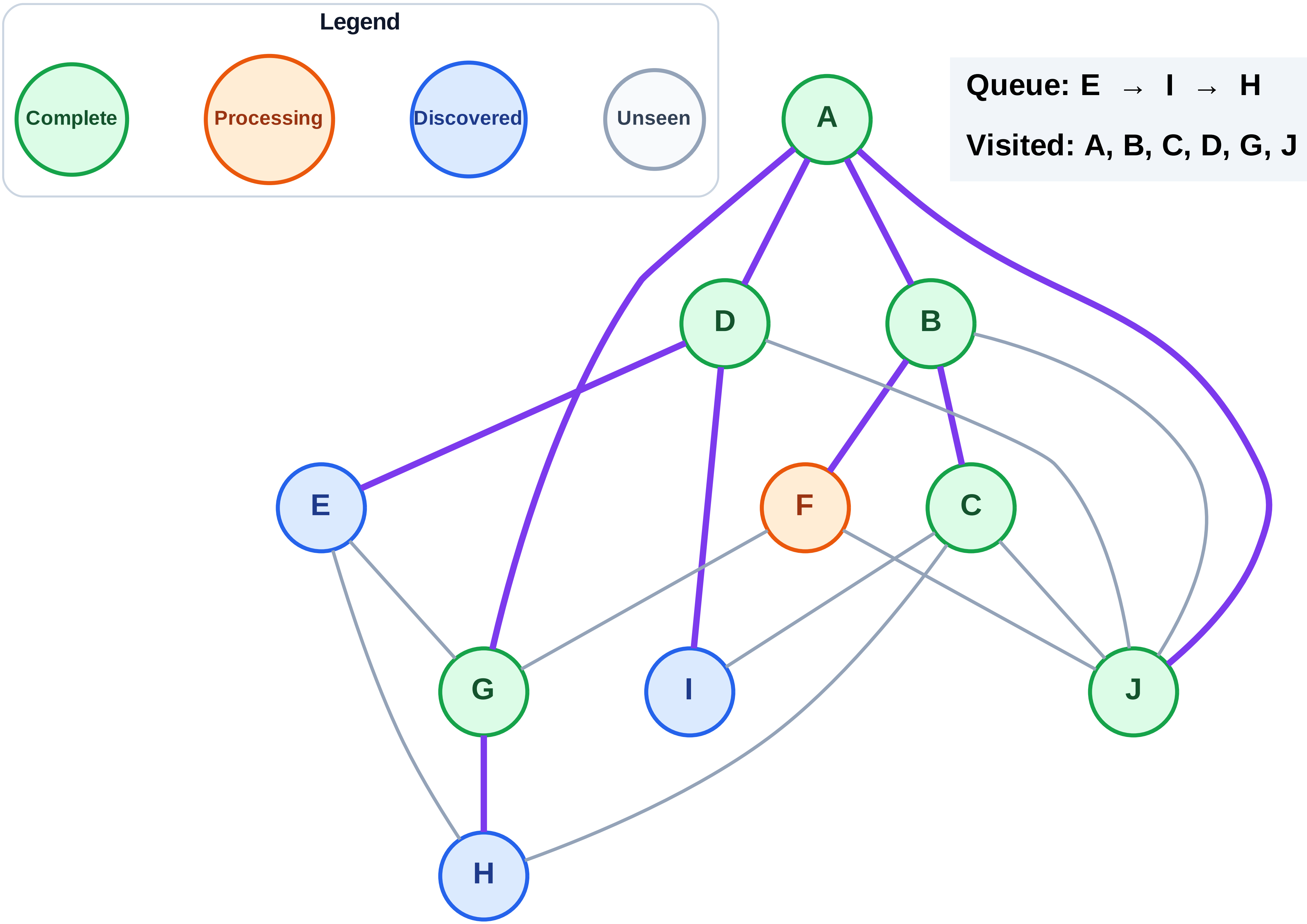
BFS Traversal

Step 13: No new neighbors from C



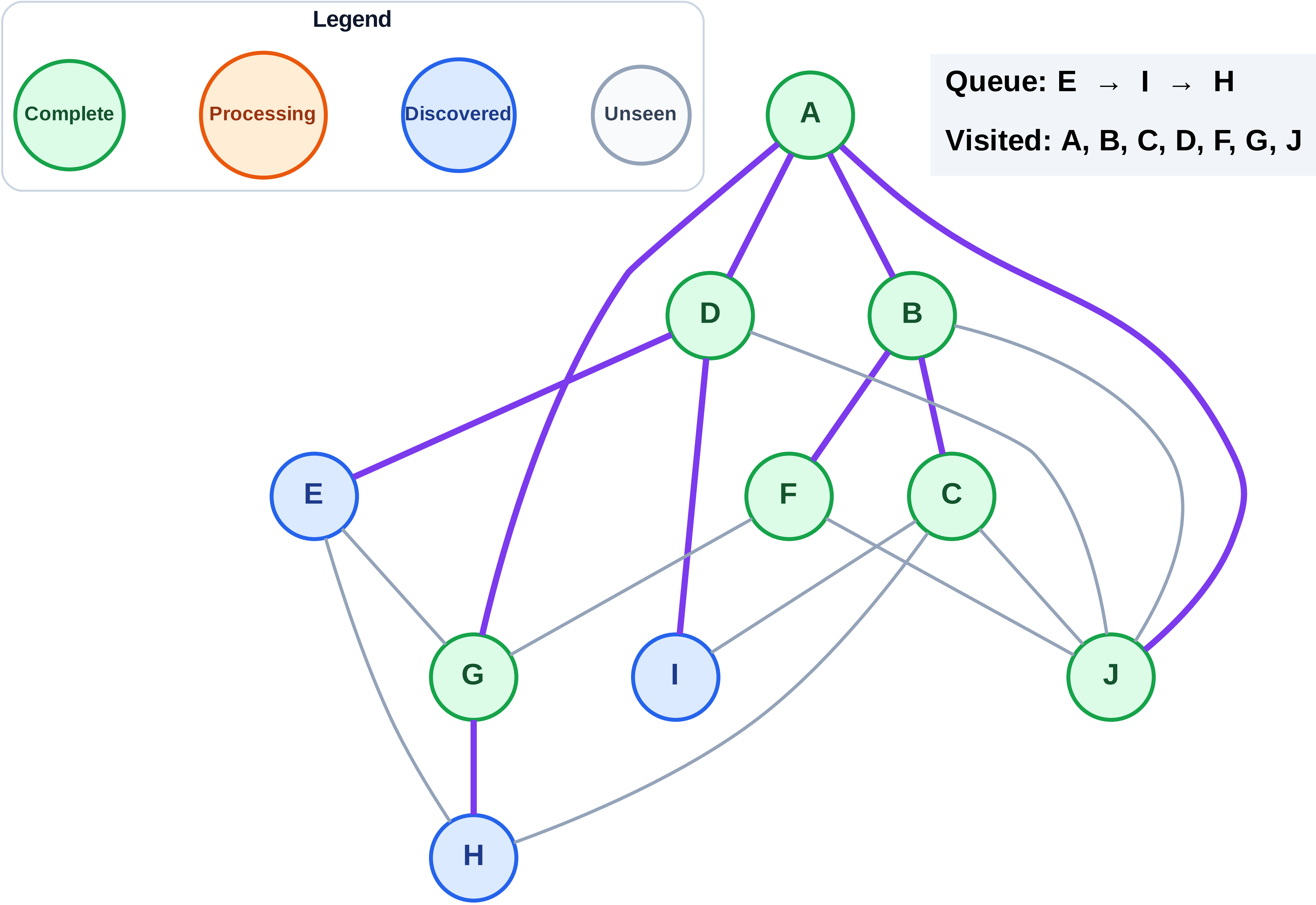
BFS Traversal

Step 14: Process node F



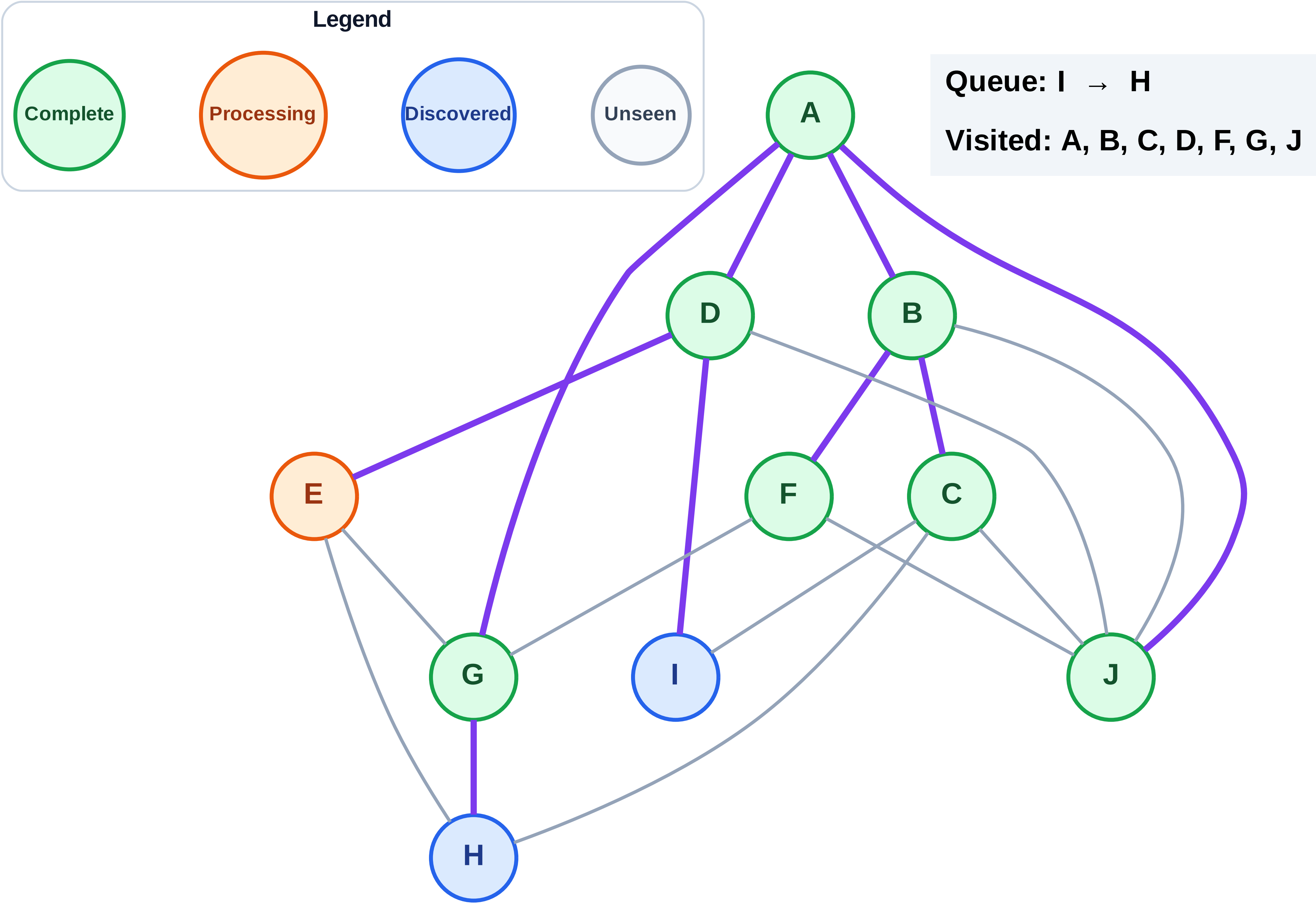
BFS Traversal

Step 15: No new neighbors from F



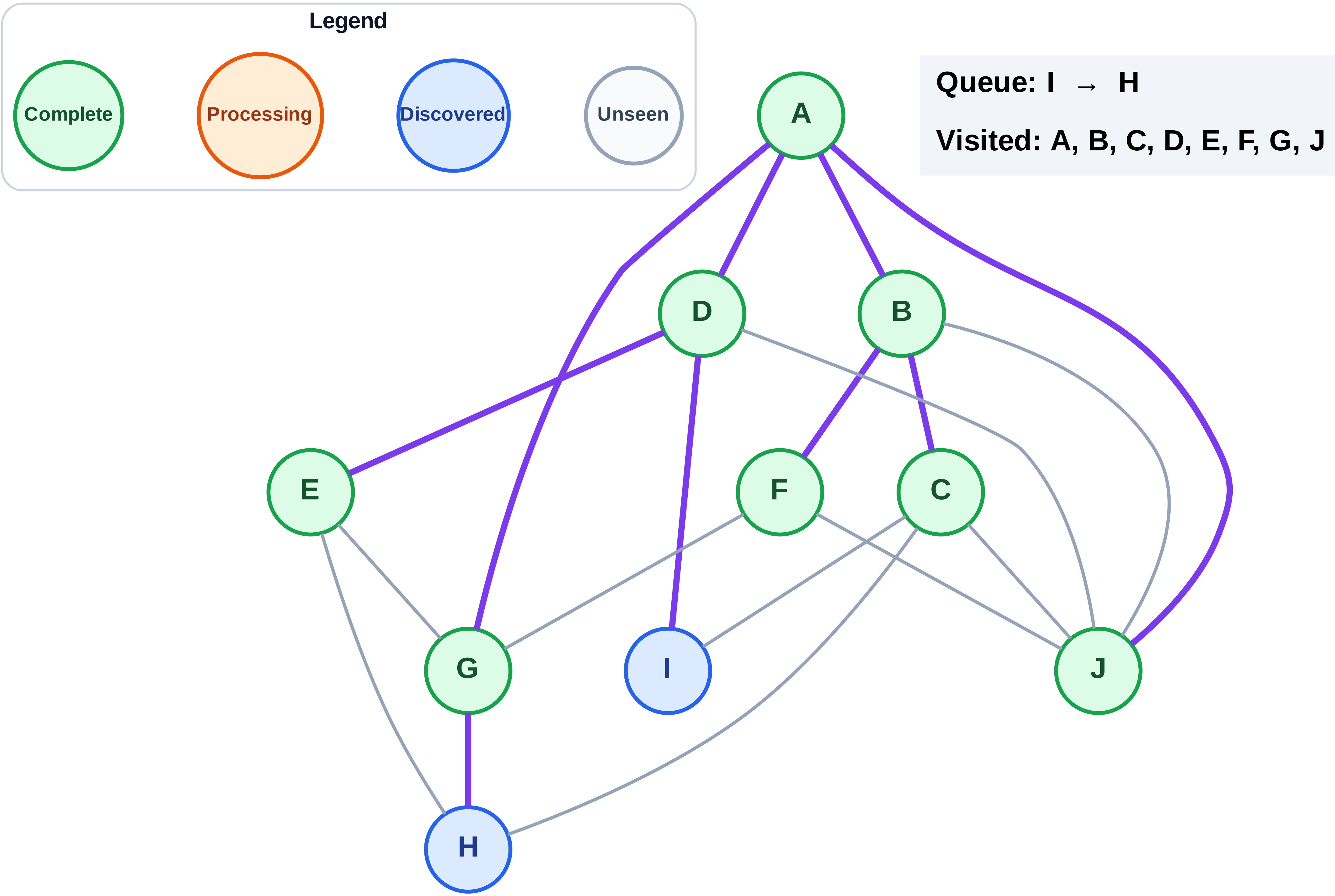
BFS Traversal

Step 16: Process node E



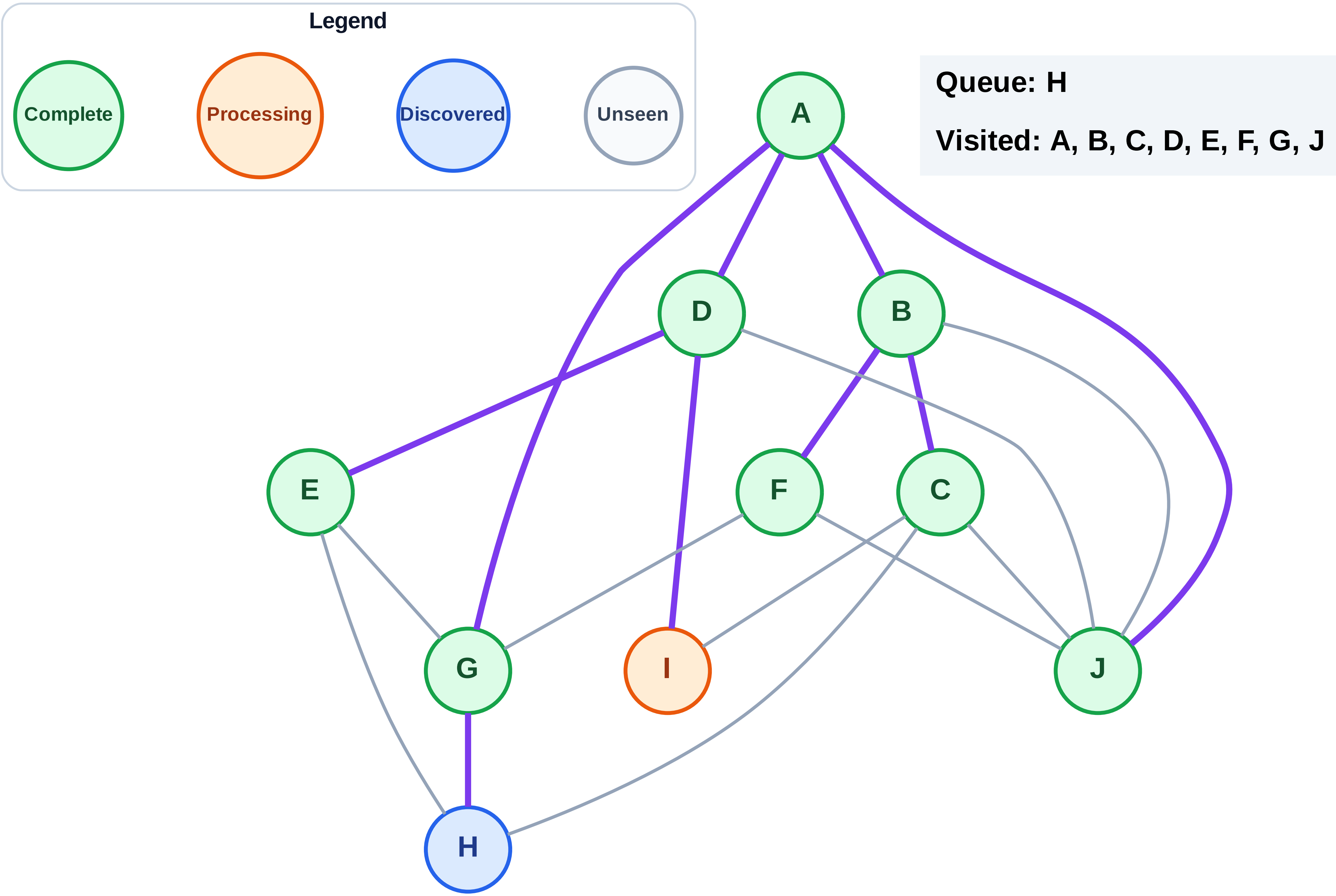
BFS Traversal

Step 17: No new neighbors from E



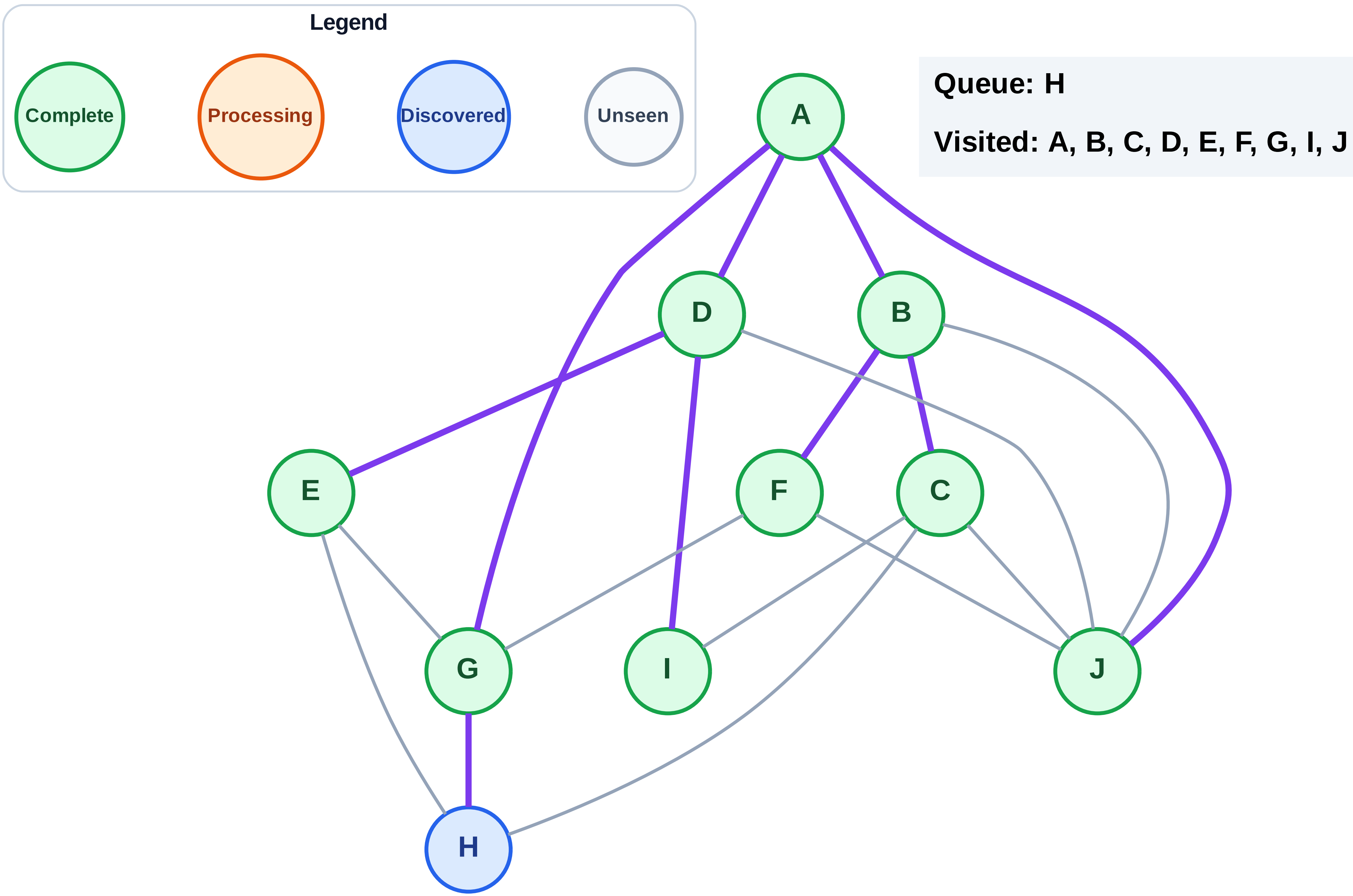
BFS Traversal

Step 18: Process node I



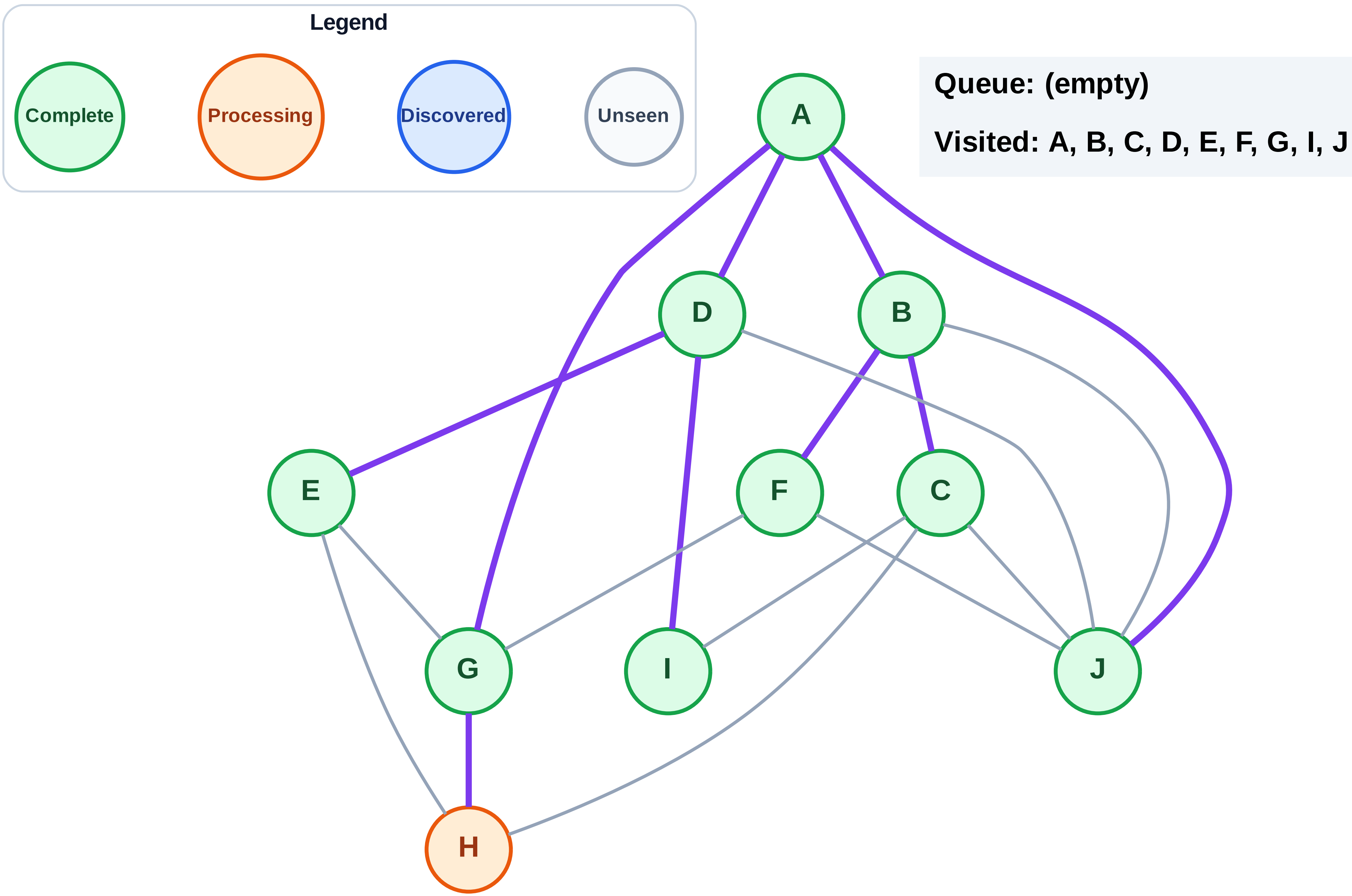
BFS Traversal

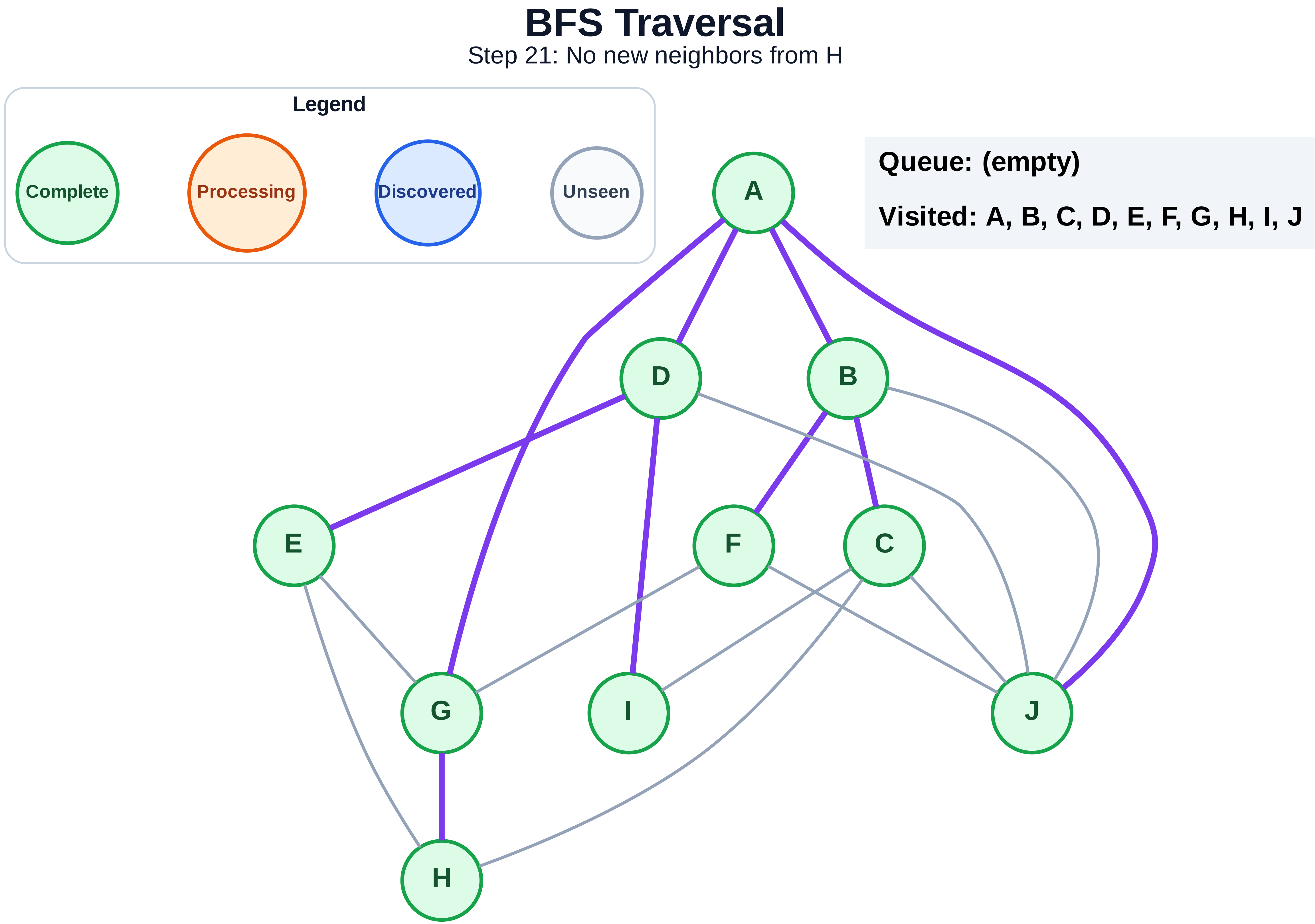
Step 19: No new neighbors from I



BFS Traversal

Step 20: Process node H




BFS Traversal
Step 21: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
D
B
E
F
C
G
I
J
H

DFS Traversal

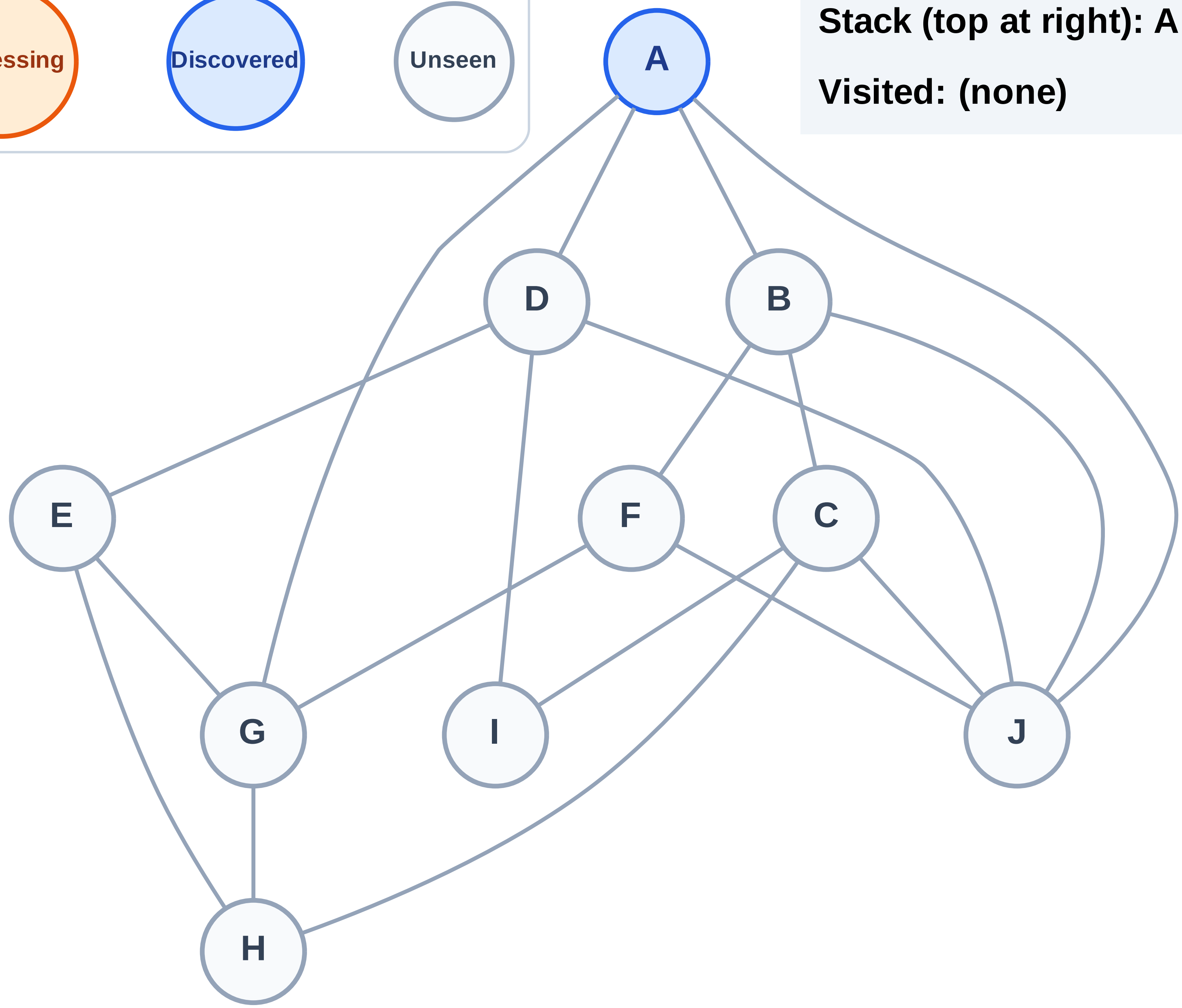
Step 1: Start at A

Legend

Complete | Processing | Discovered | Unseen

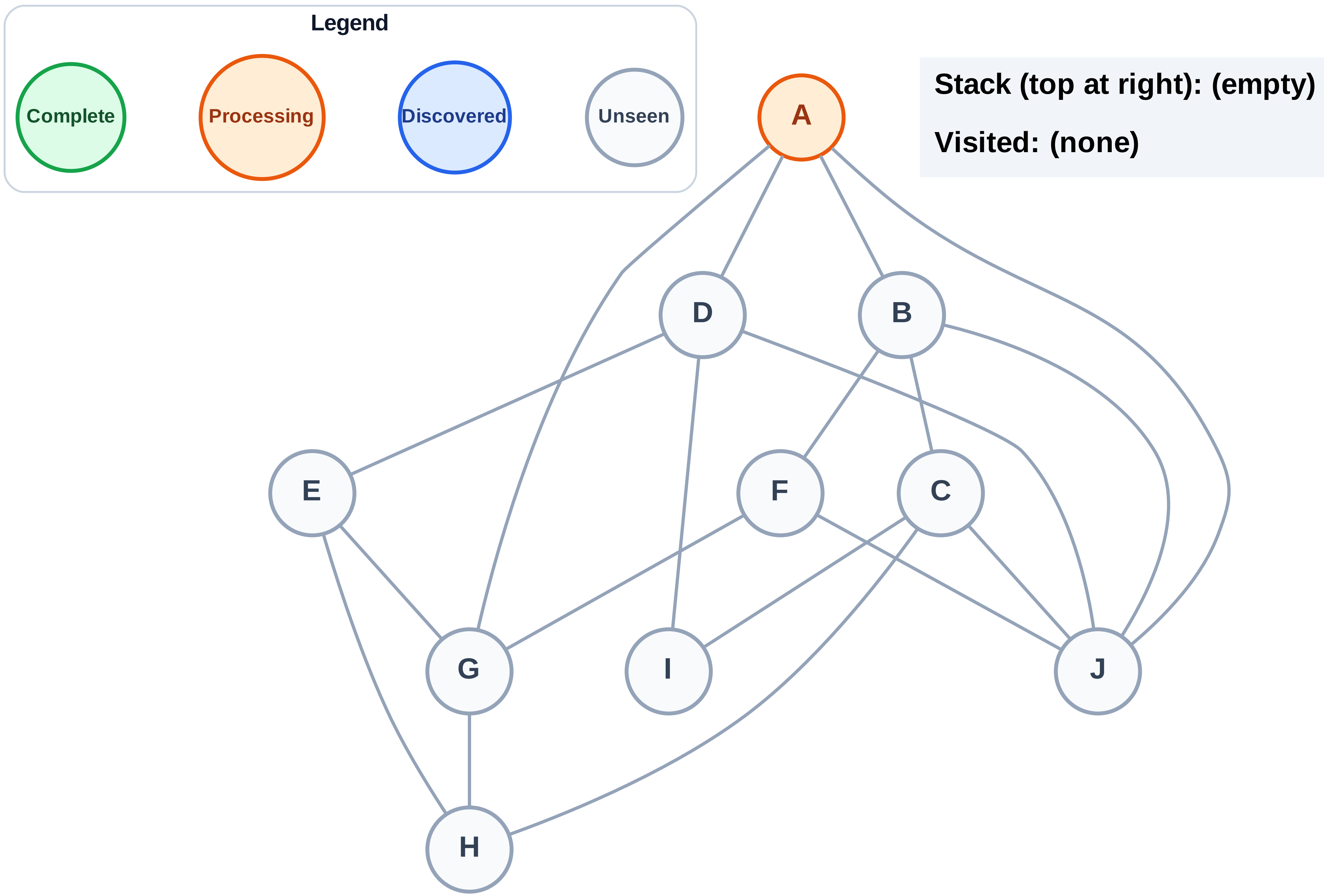
Stack (top at right): A

Visited: (none)



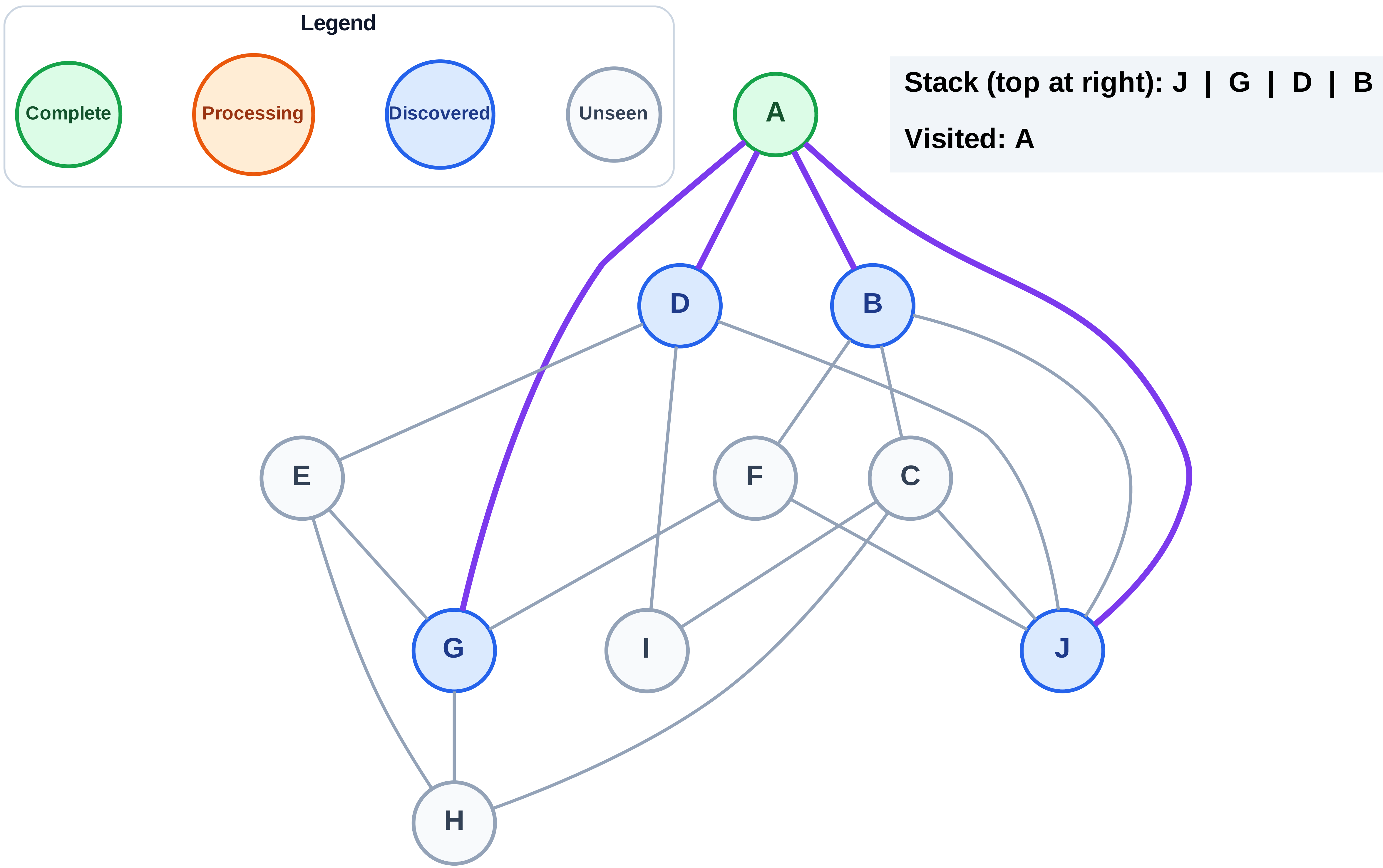
DFS Traversal

Step 2: Process node A



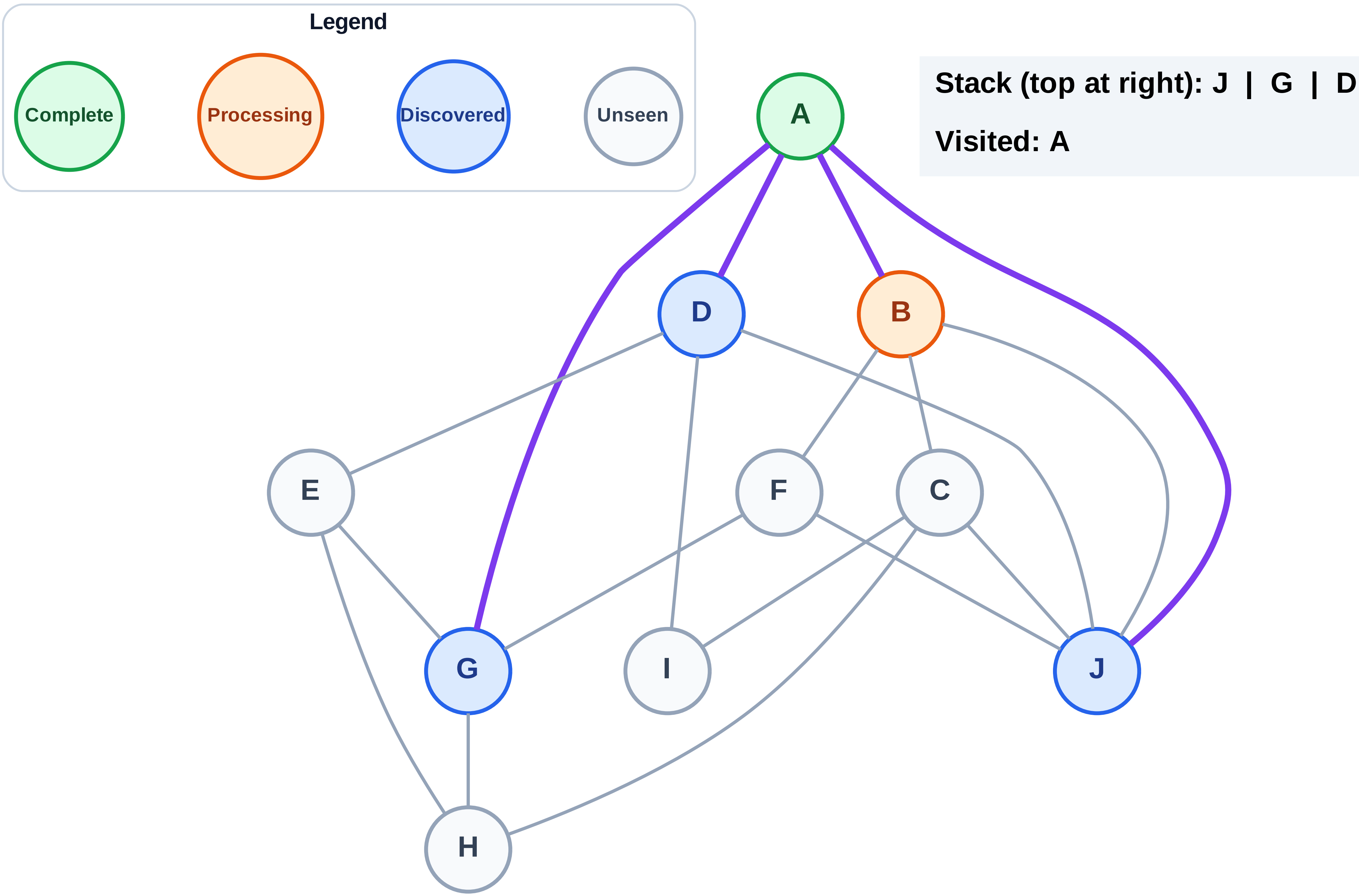
DFS Traversal

Step 3: Discover J, G, D, B from A



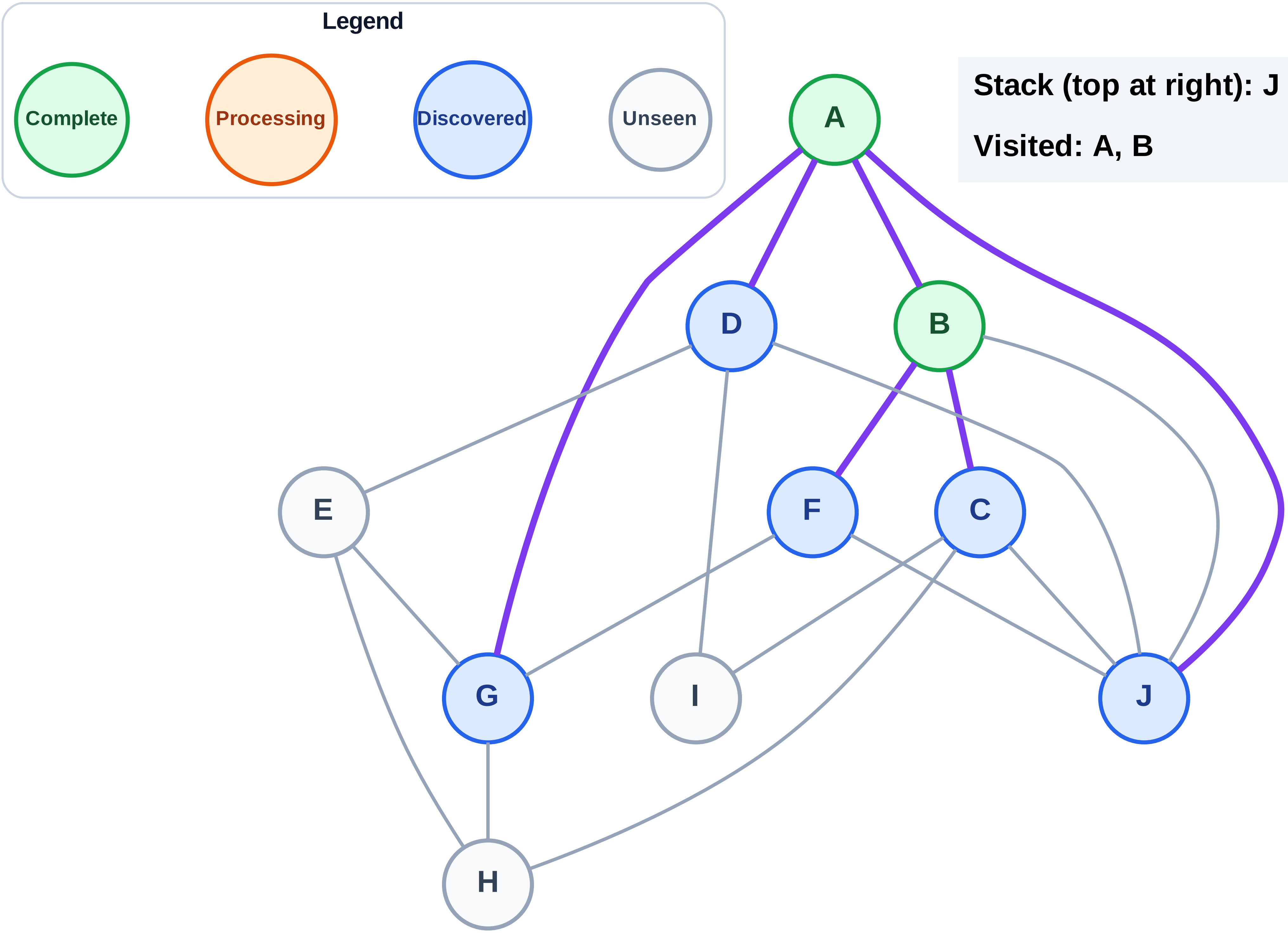
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover F, C from B

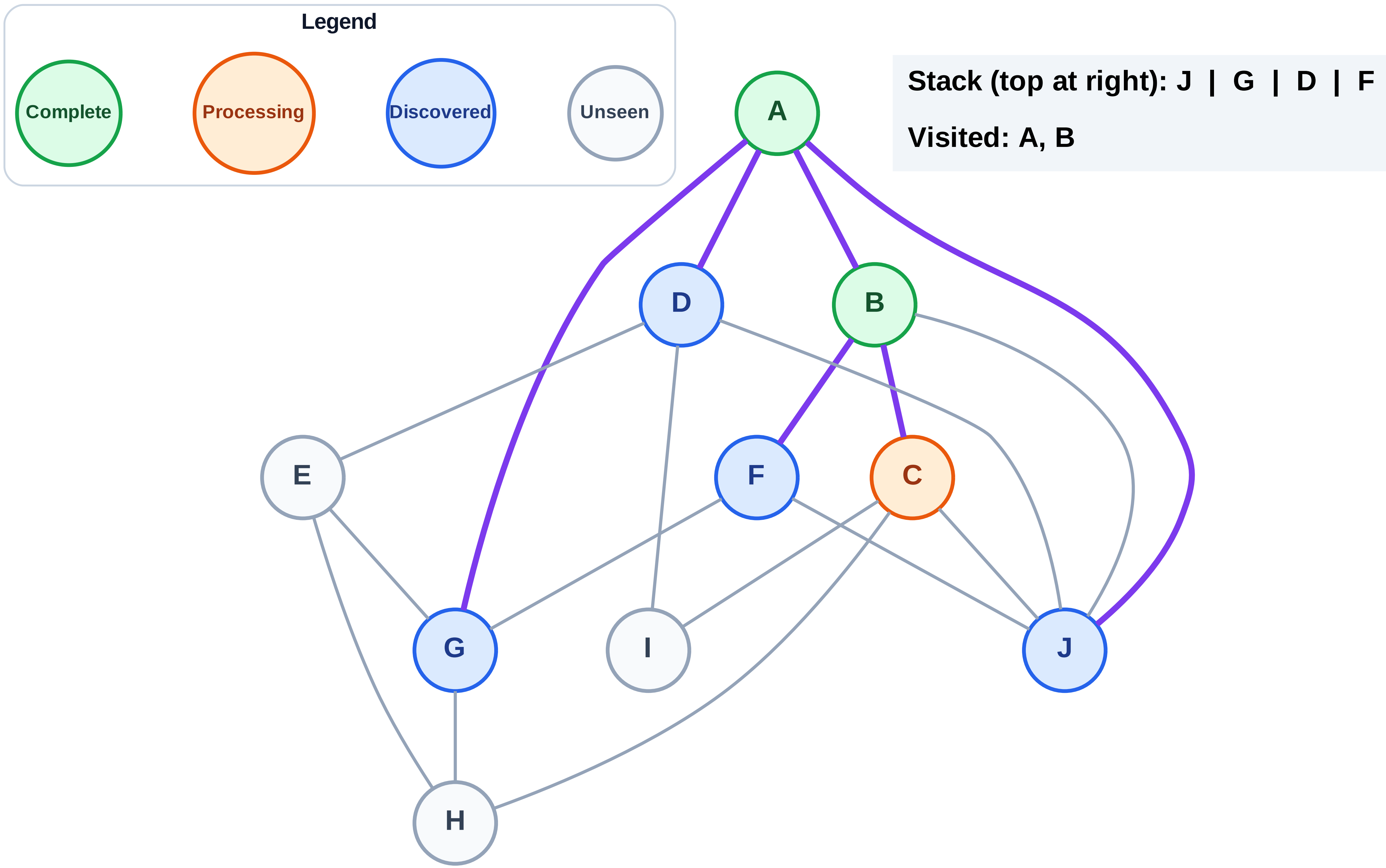


Stack (top at right): J | G | D | F | C

Visited: A, B

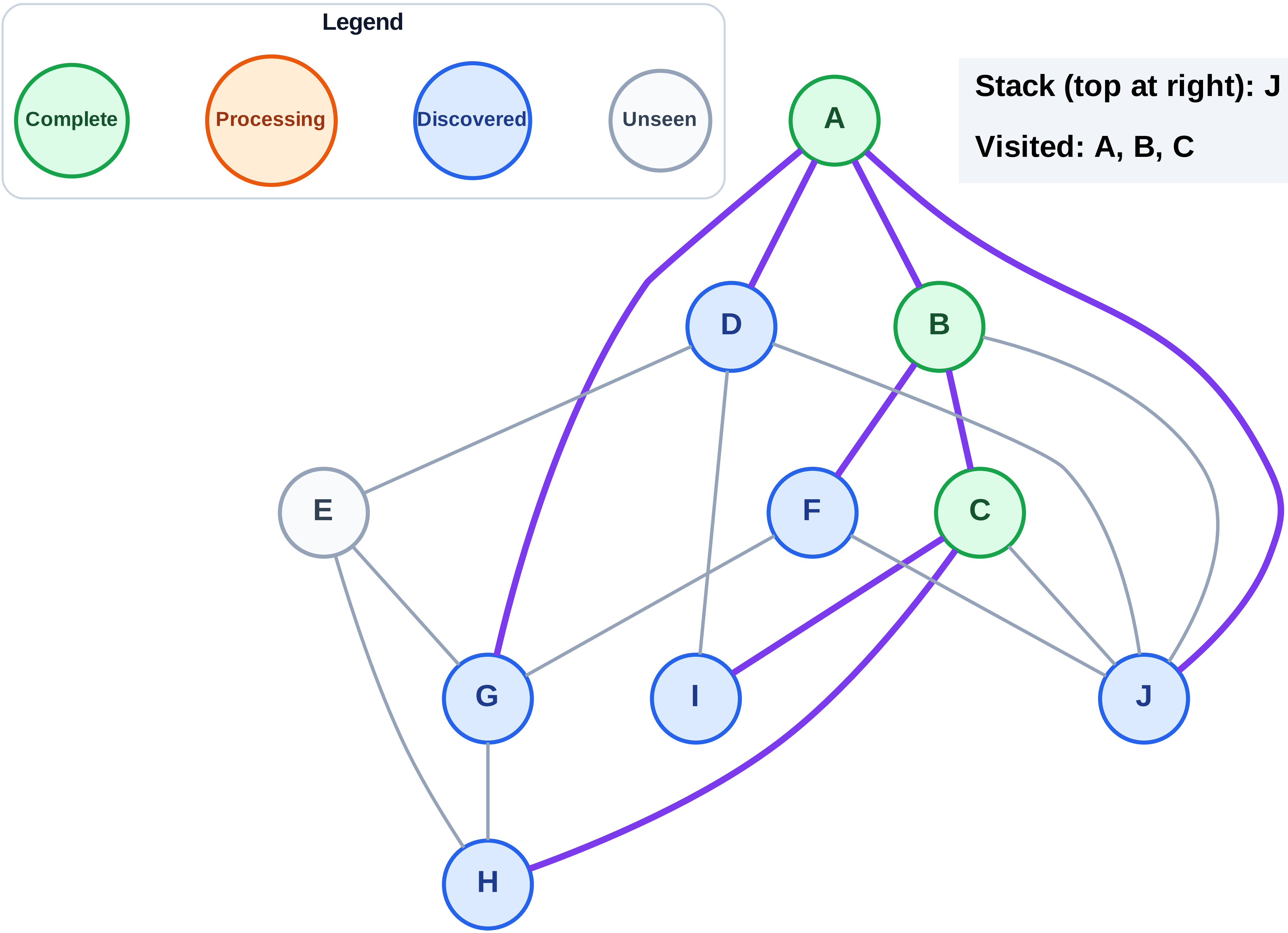
DFS Traversal

Step 6: Process node C



DFS Traversal

Step 7: Discover I, H from C

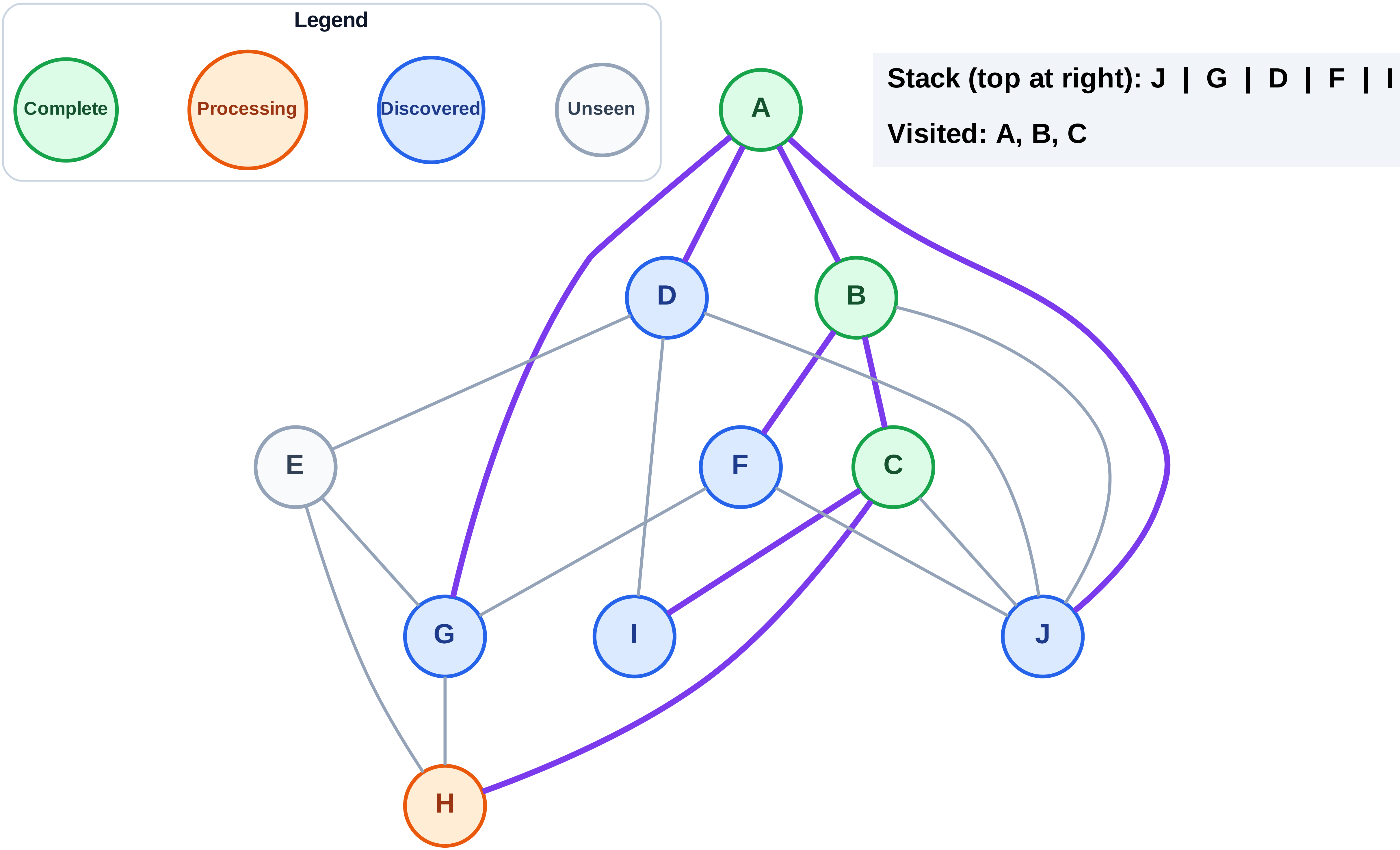


Stack (top at right): J | G | D | F | I | H

Visited: A, B, C

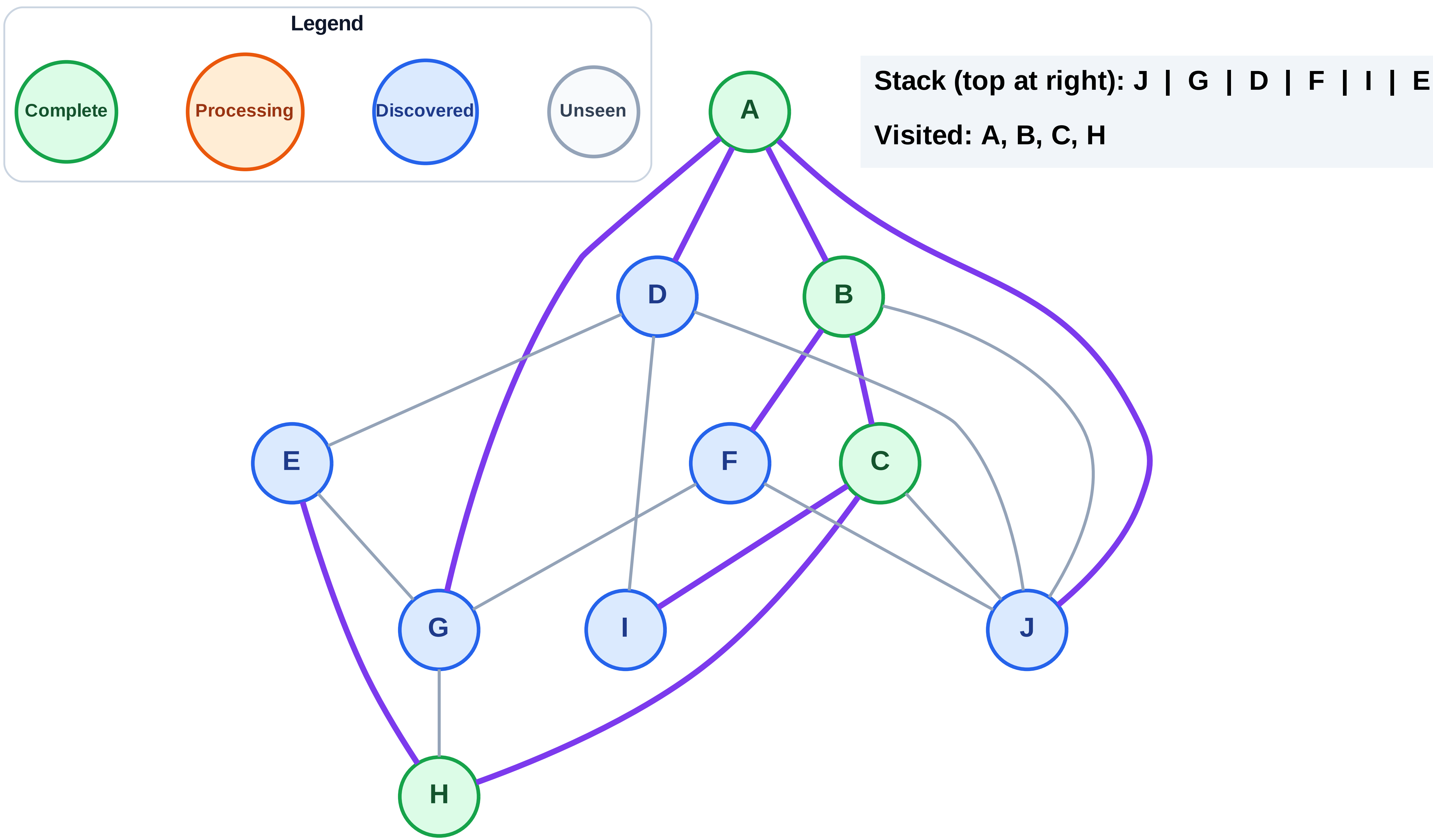
DFS Traversal

Step 8: Process node H



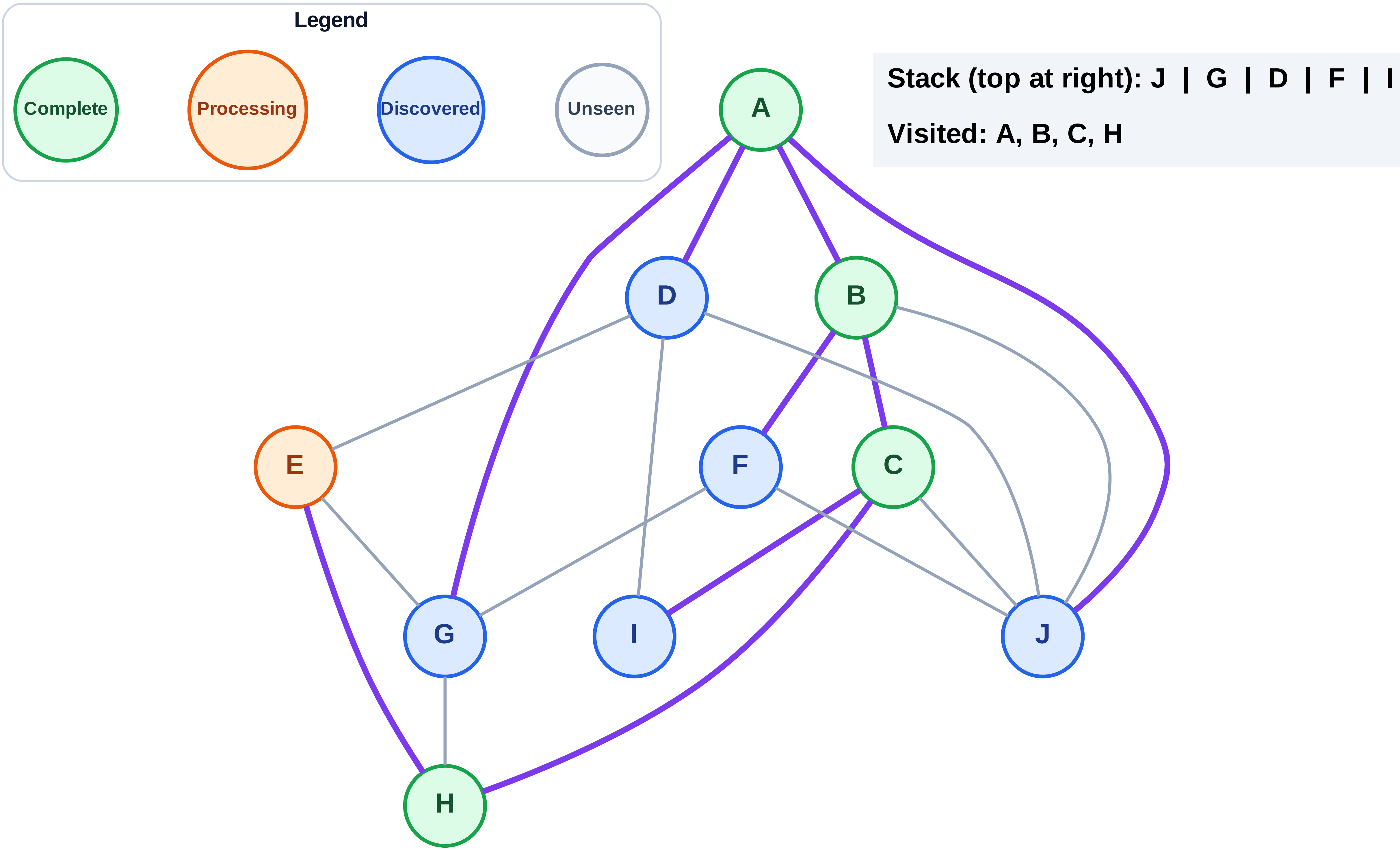
DFS Traversal

Step 9: Discover E from H



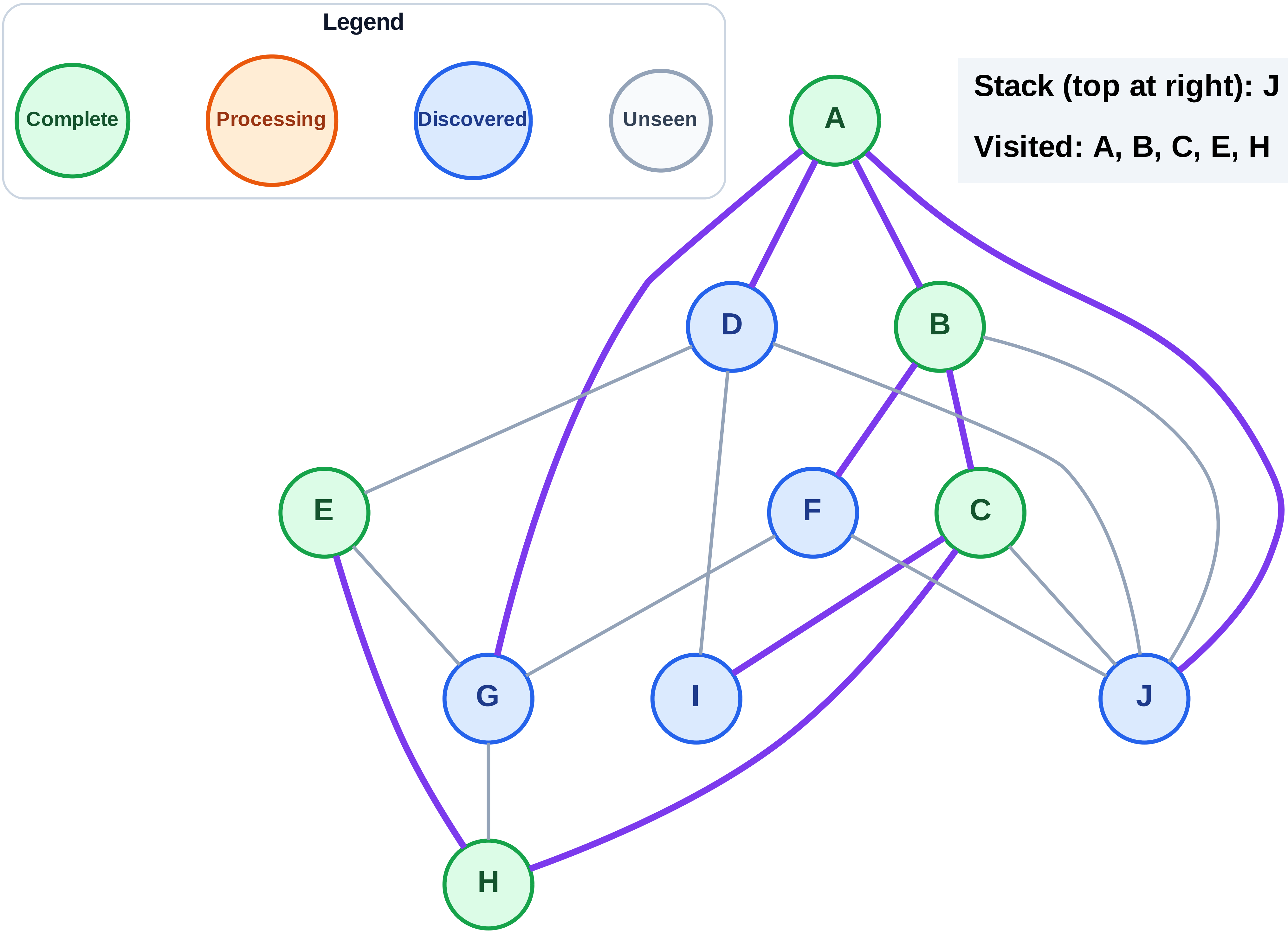
DFS Traversal

Step 10: Process node E



DFS Traversal

Step 11: No new neighbors from E

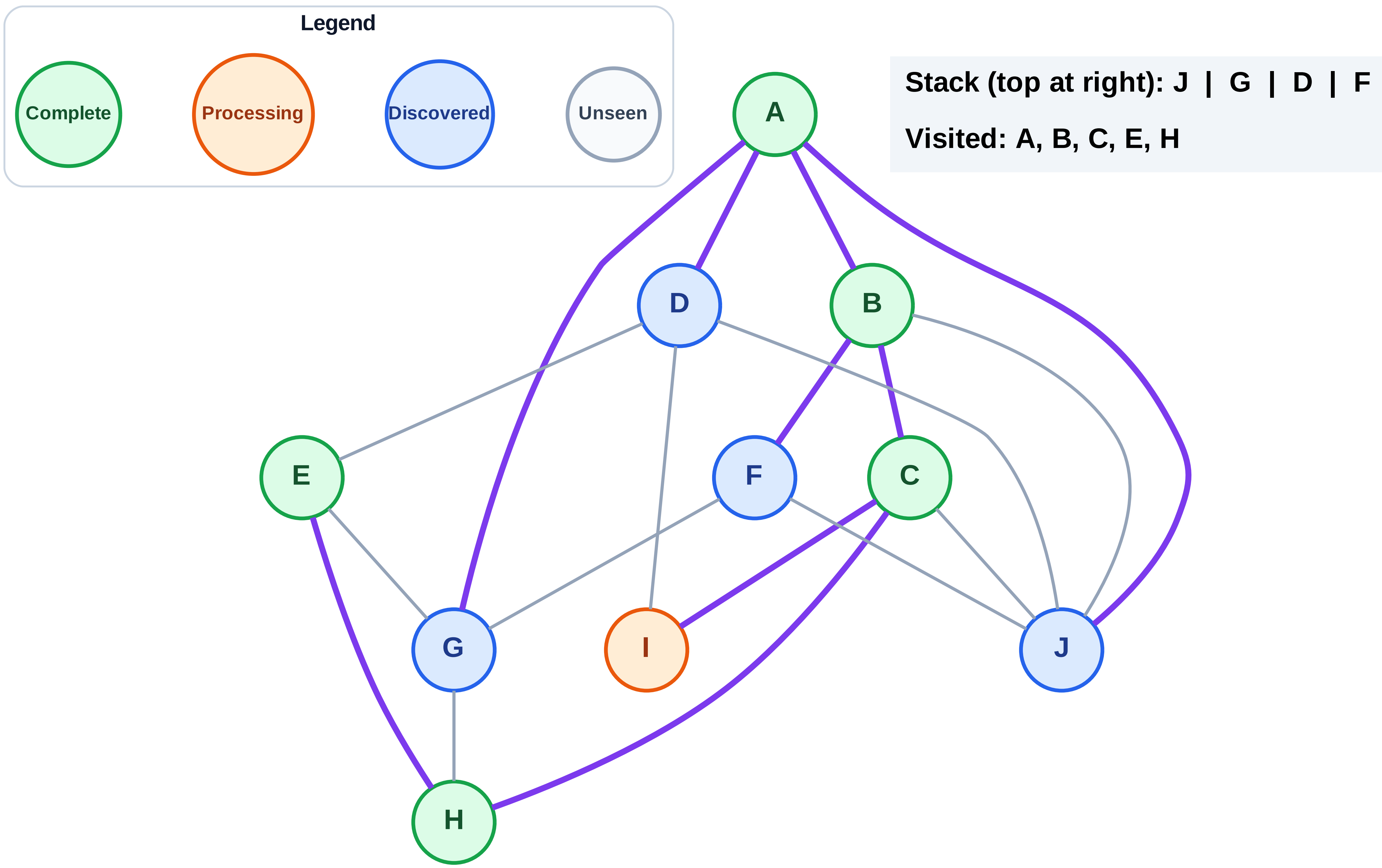


Stack (top at right): J | G | D | F | I

Visited: A, B, C, E, H

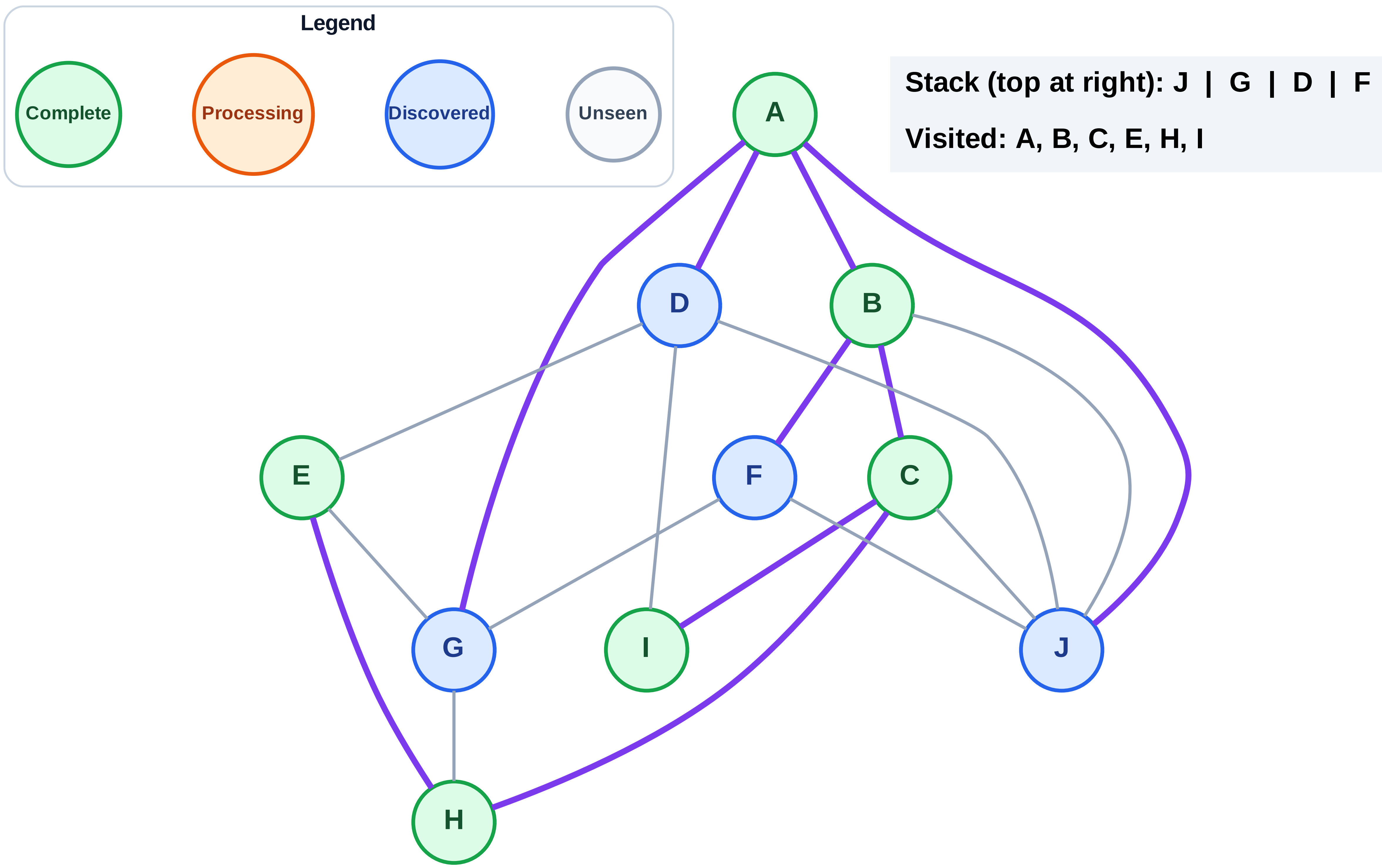
DFS Traversal

Step 12: Process node I



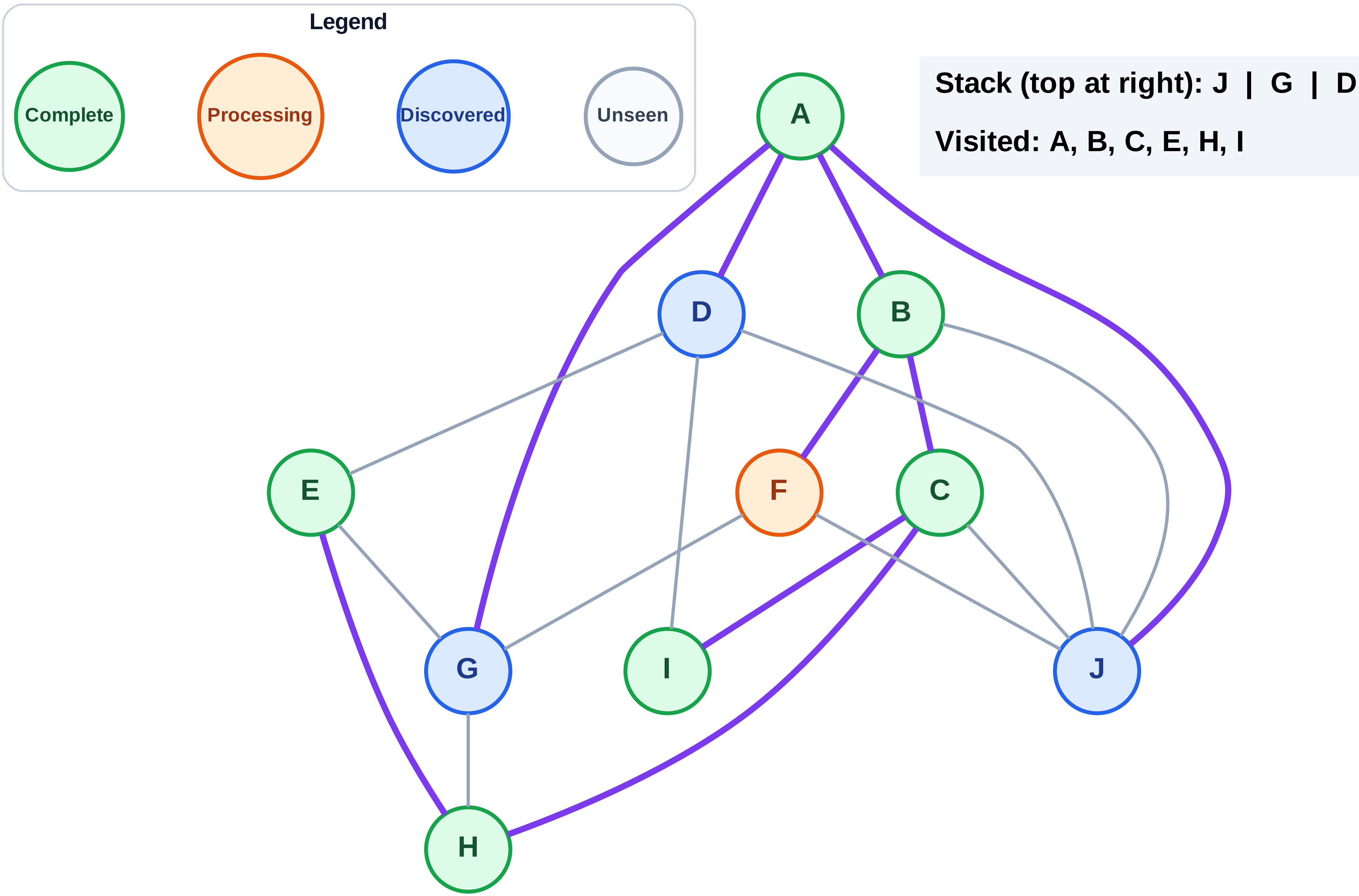
DFS Traversal

Step 13: No new neighbors from I



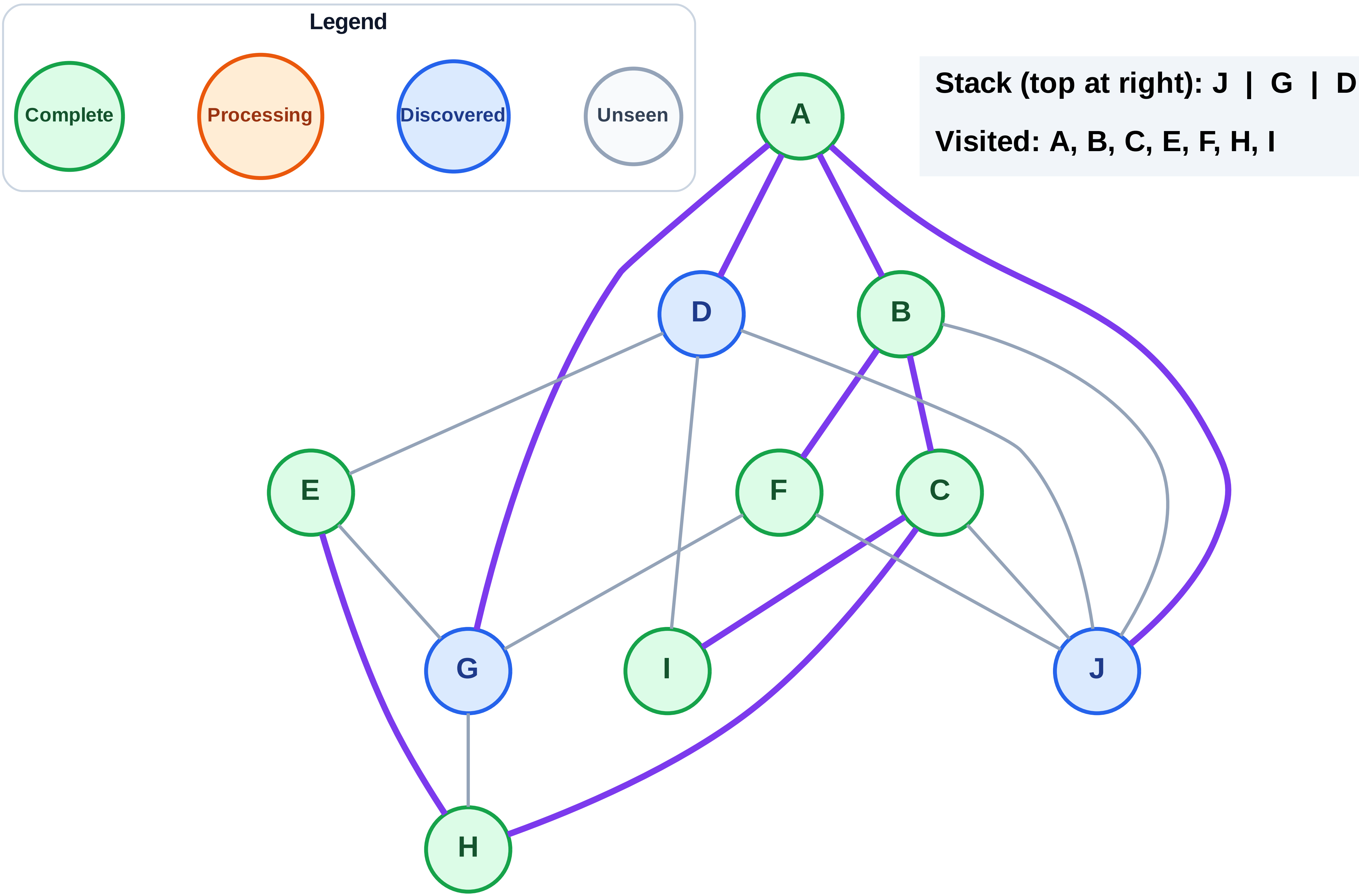
DFS Traversal

Step 14: Process node F



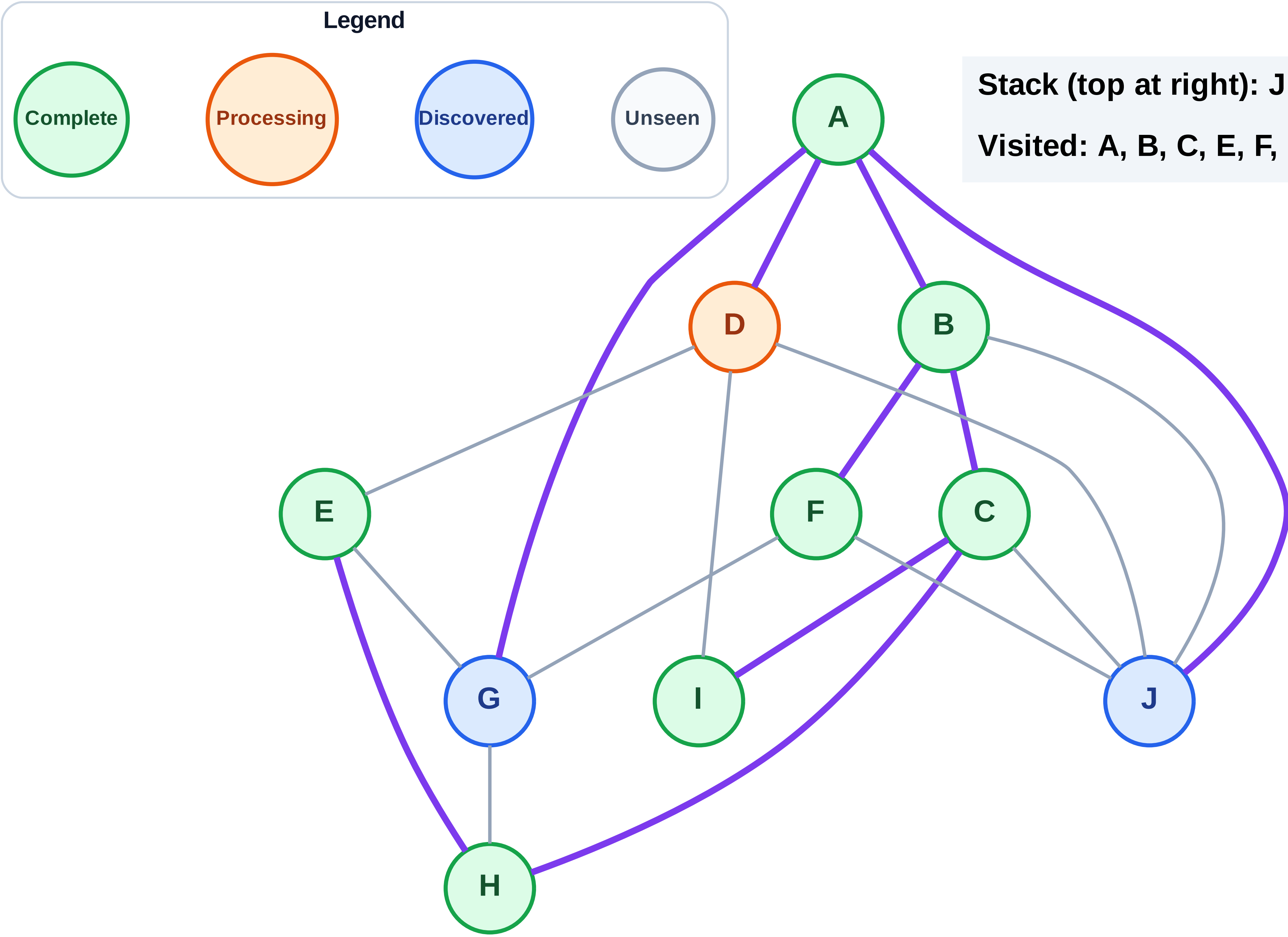
DFS Traversal

Step 15: No new neighbors from F



DFS Traversal

Step 16: Process node D

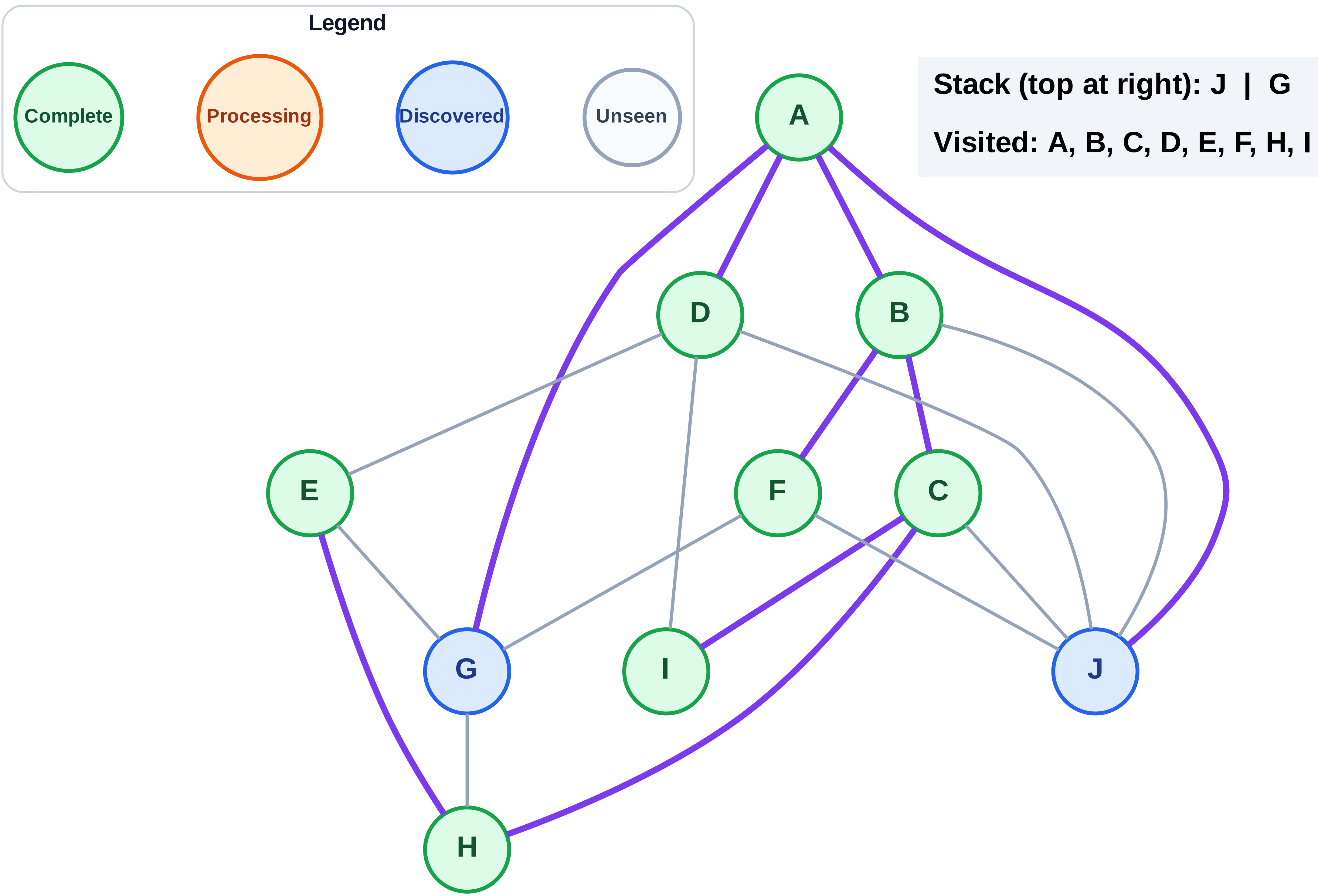


Stack (top at right): J | G

Visited: A, B, C, E, F, H, I

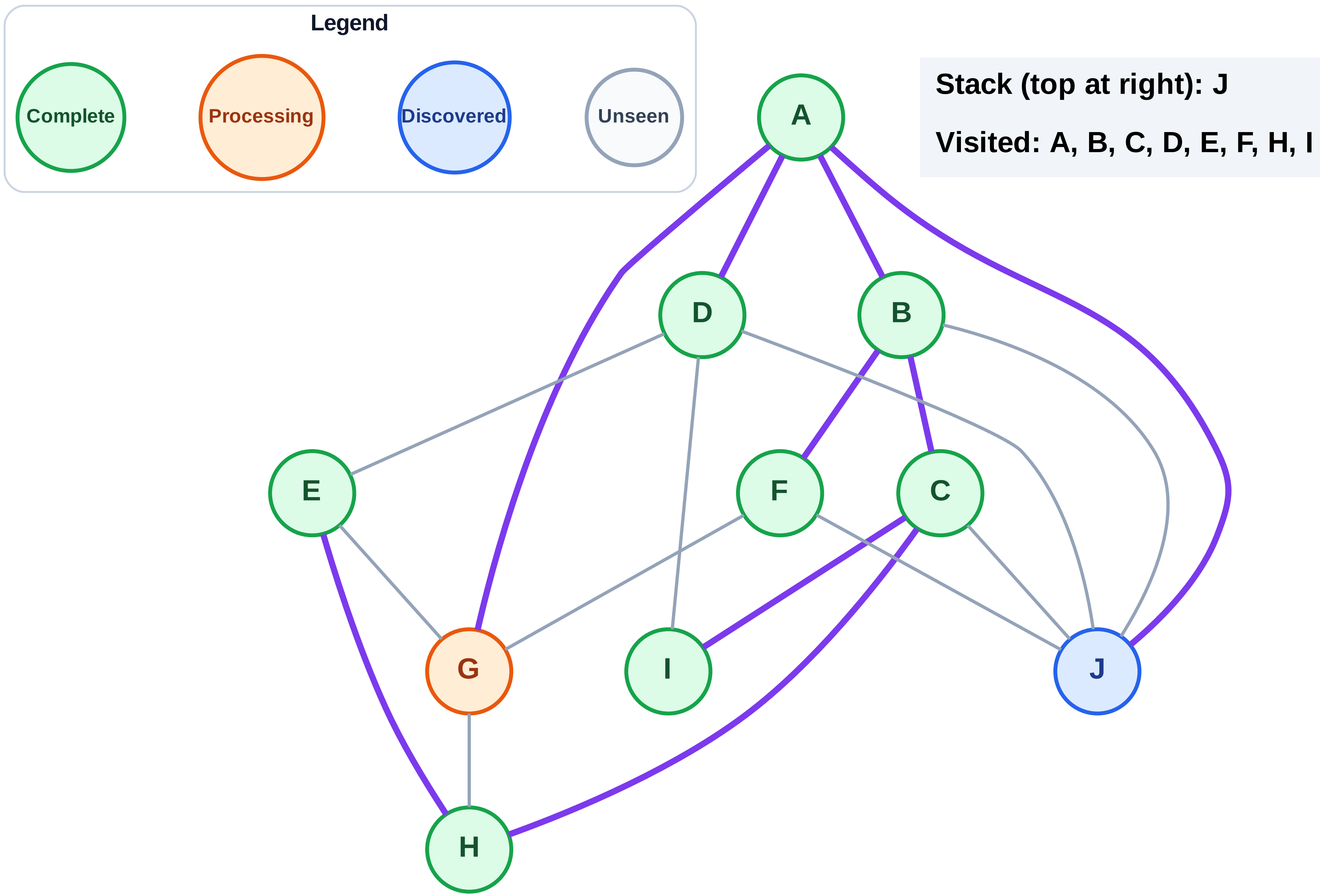
DFS Traversal

Step 17: No new neighbors from D



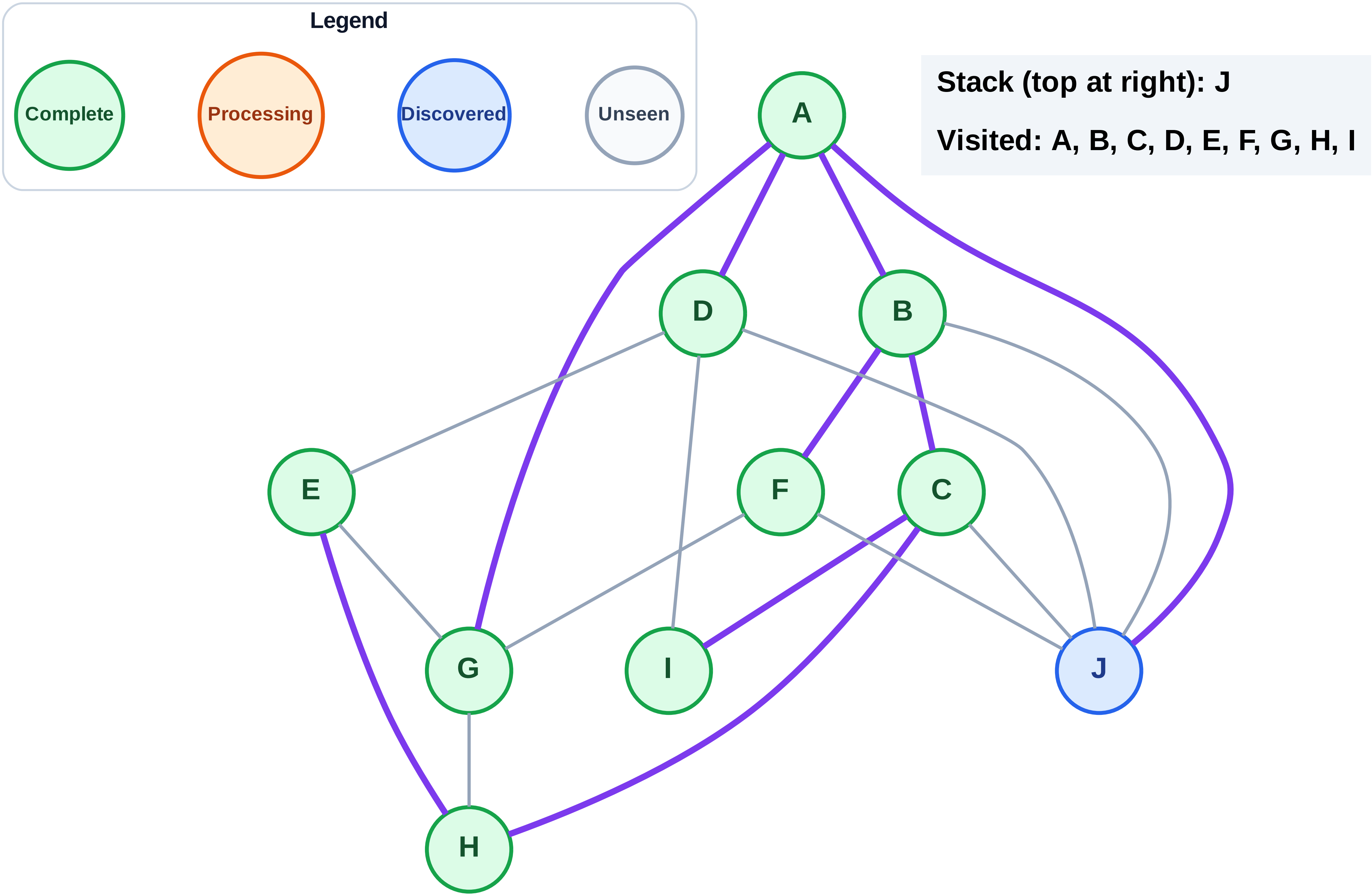
DFS Traversal

Step 18: Process node G



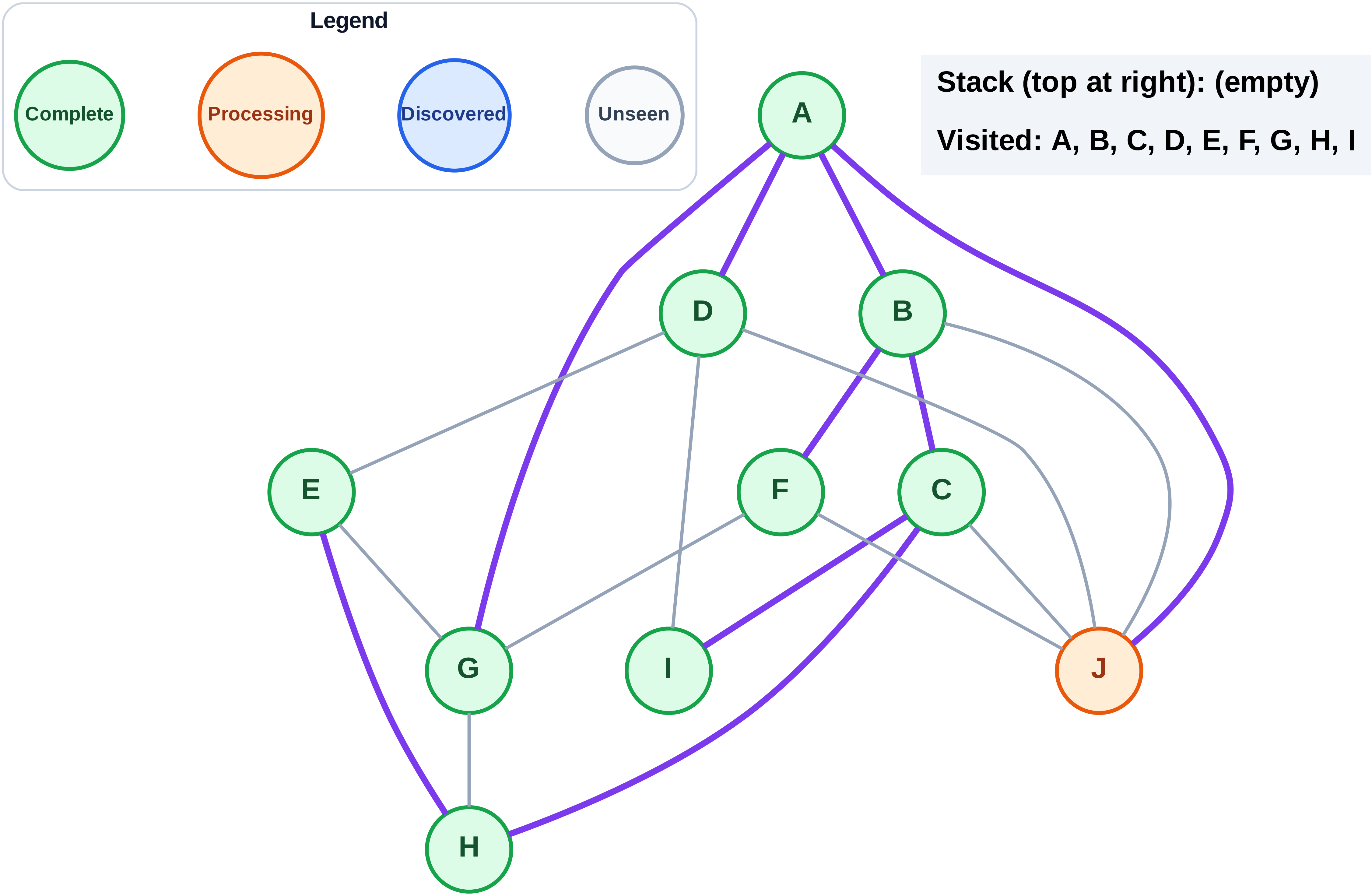
DFS Traversal

Step 19: No new neighbors from G



DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J